

Florida Keys National Marine Sanctuary

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

Contents

Funding & Acknowledgements	2
Threshold Filtering	2
Value Qualifiers	3
Water Column	5
Seasonal Kendall-Tau Analysis	5
Water Quality - Discrete	5
Chlorophyll a, Corrected for Pheophytin - Discrete	6
Chlorophyll a, Uncorrected for Pheophytin - Discrete	7
Colored Dissolved Organic Matter - Discrete	10
Dissolved Oxygen - Discrete	11
Dissolved Oxygen Saturation - Discrete	14
pH - Discrete	17
Salinity - Discrete	19
Total Nitrogen - Discrete	21
Total Phosphorus - Discrete	24
Turbidity - Discrete	26
Water Temperature - Discrete	28
Water Quality - Continuous	31
pH - Continuous	34
Dissolved Oxygen Saturation - Continuous	35
Dissolved Oxygen - Continuous	36
Turbidity - Continuous	37
Water Temperature - Continuous - Program 2	38
Water Temperature - Continuous - Program 5	39
Water Temperature - Continuous - Program 7	40
Water Temperature - Continuous - Program 296	41
Water Temperature - Continuous - Program 899	43
Water Temperature - Continuous - Program 986	44
Water Temperature - Continuous - Program 989	46
Water Temperature - Continuous - Program 10004	47
Salinity - Continuous	49
Specific Conductivity - Continuous	50
Submerged Aquatic Vegetation	51
Parameters	51
Species	51
Notes	51
SAV Water Column Analysis	56

Coral Reef	58
Species list	62
References	67

Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

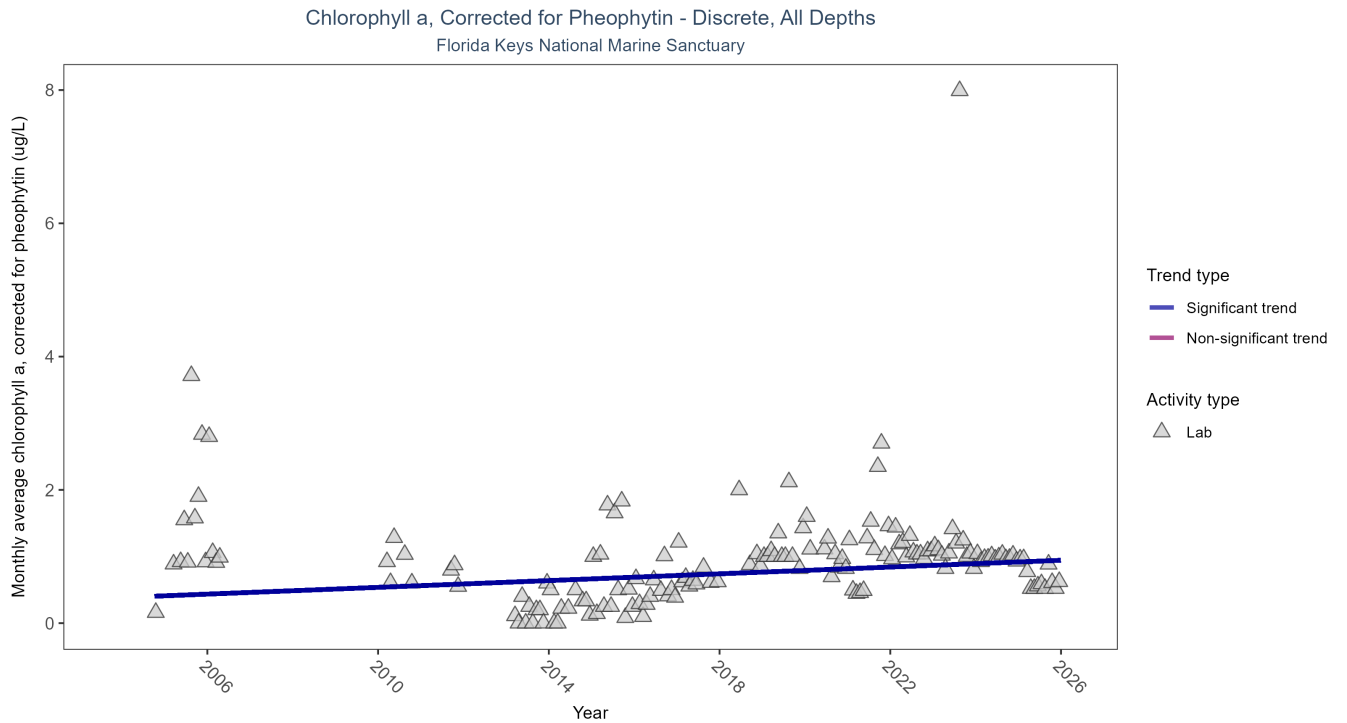


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2204	18	2004 - 2025	0.6725	0.1896	0.3848	0.0254	0.0011

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.03 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

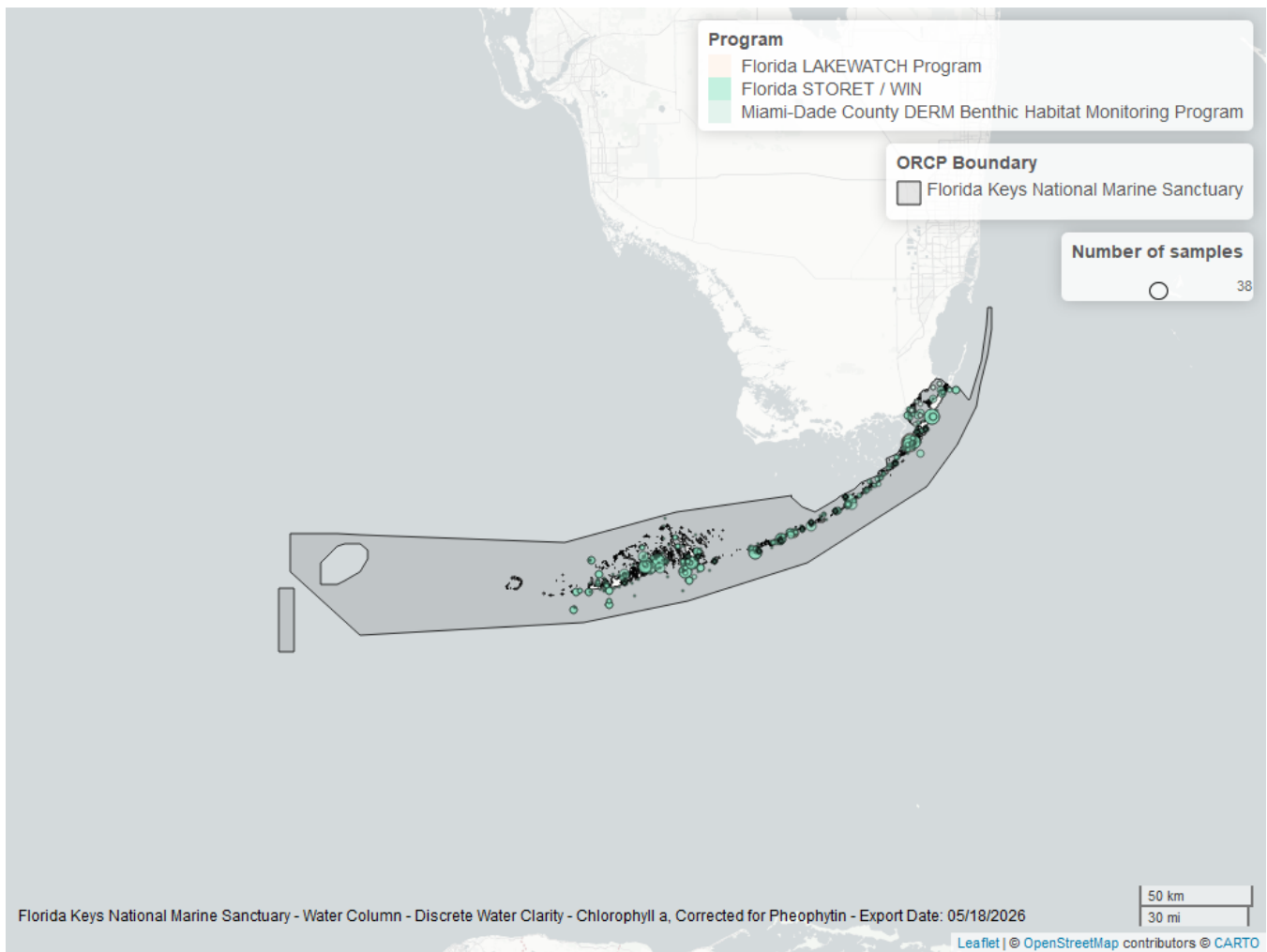


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	2004	2005	2025
514	273	2018	2025
4018	26	2004	2011

Program names:

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

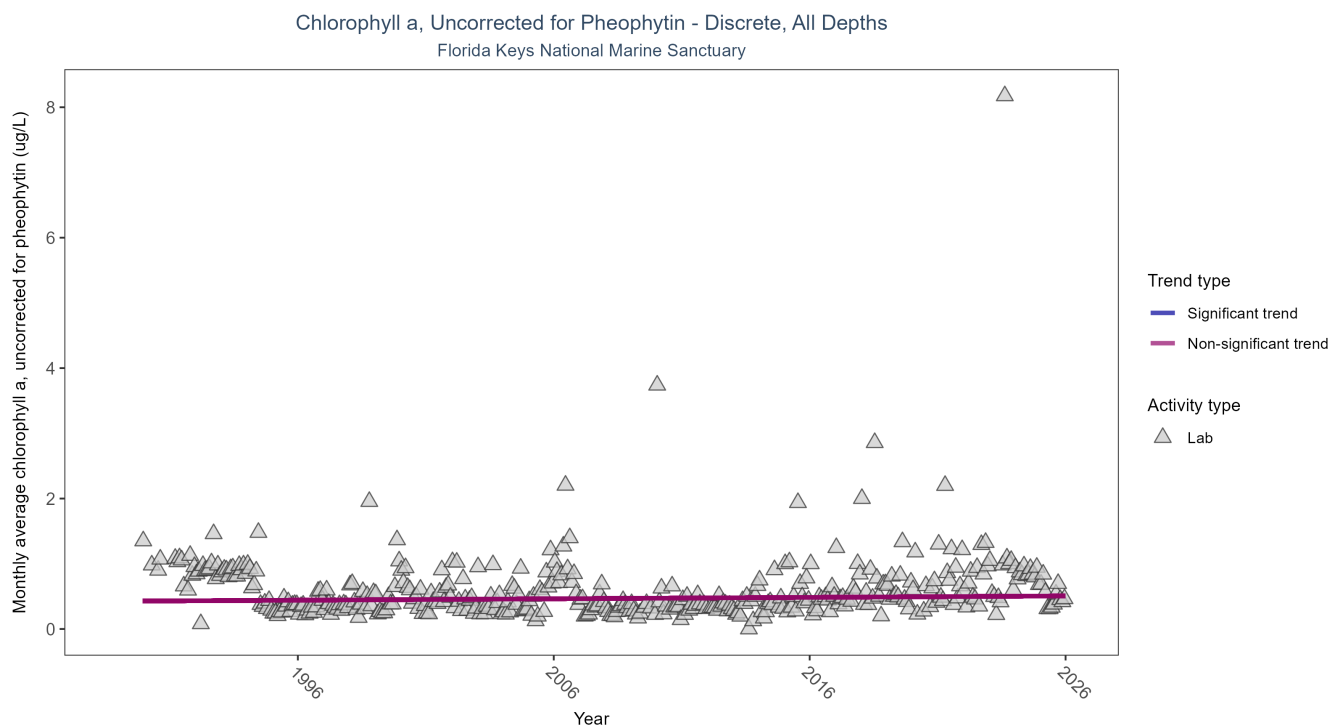


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	21260	37	1989 - 2025	0.2994	0.0613	0.4261	0.0022	0.0822

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1989 and 2025.

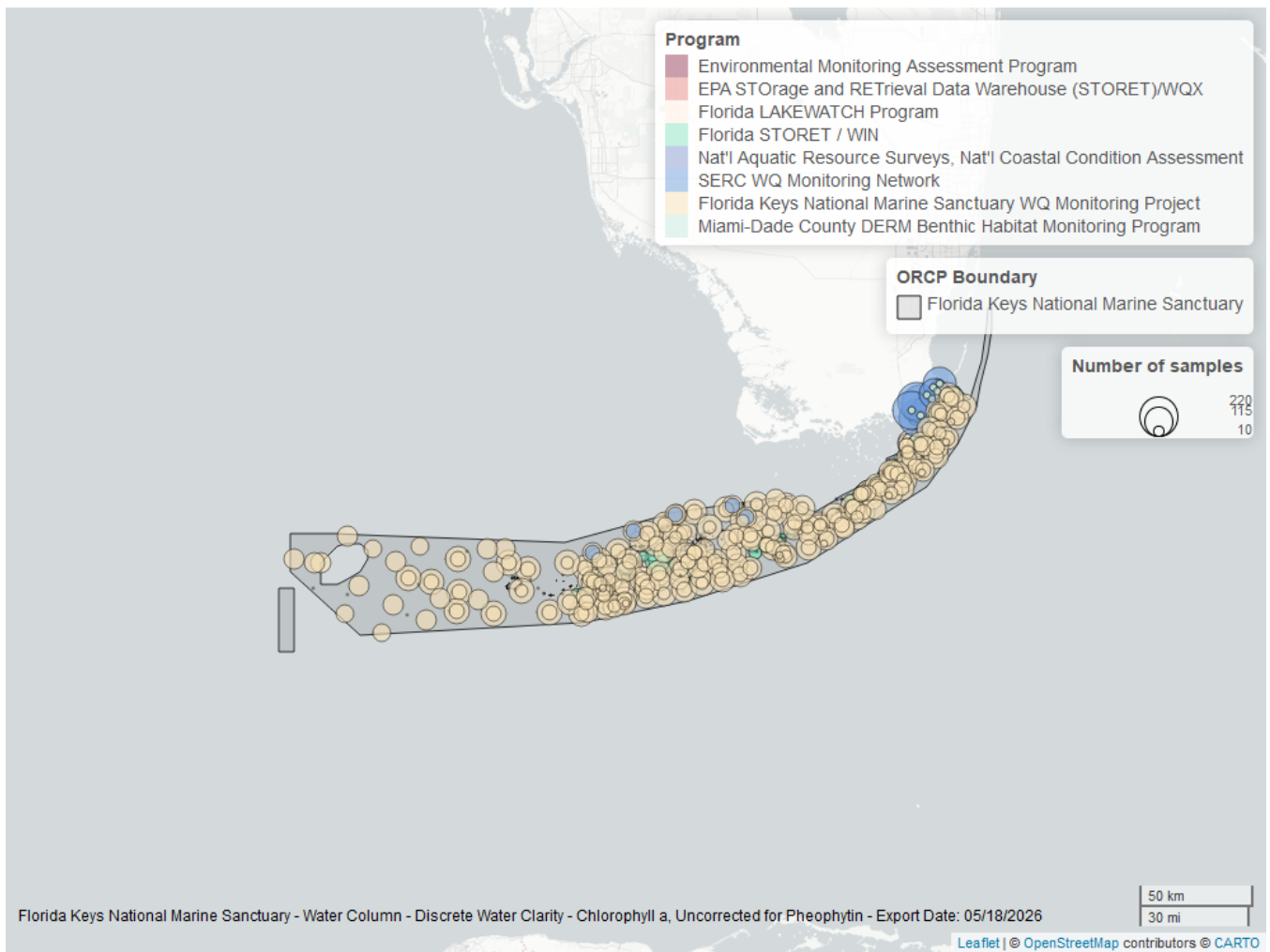


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
297	15808	1995	2023
514	2897	1998	2025
509	1336	1989	2008
5002	1074	2001	2025
103	58	2000	2015
4018	49	2025	2025
118	28	2000	2010
115	28	2000	2004

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

514 - Florida LAKEWATCH Program²

509 - SERC Water Quality Monitoring Network⁵

5002 - Florida STORET / WIN¹

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶
 4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³
 118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷
 115 - Environmental Monitoring Assessment Program⁸

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

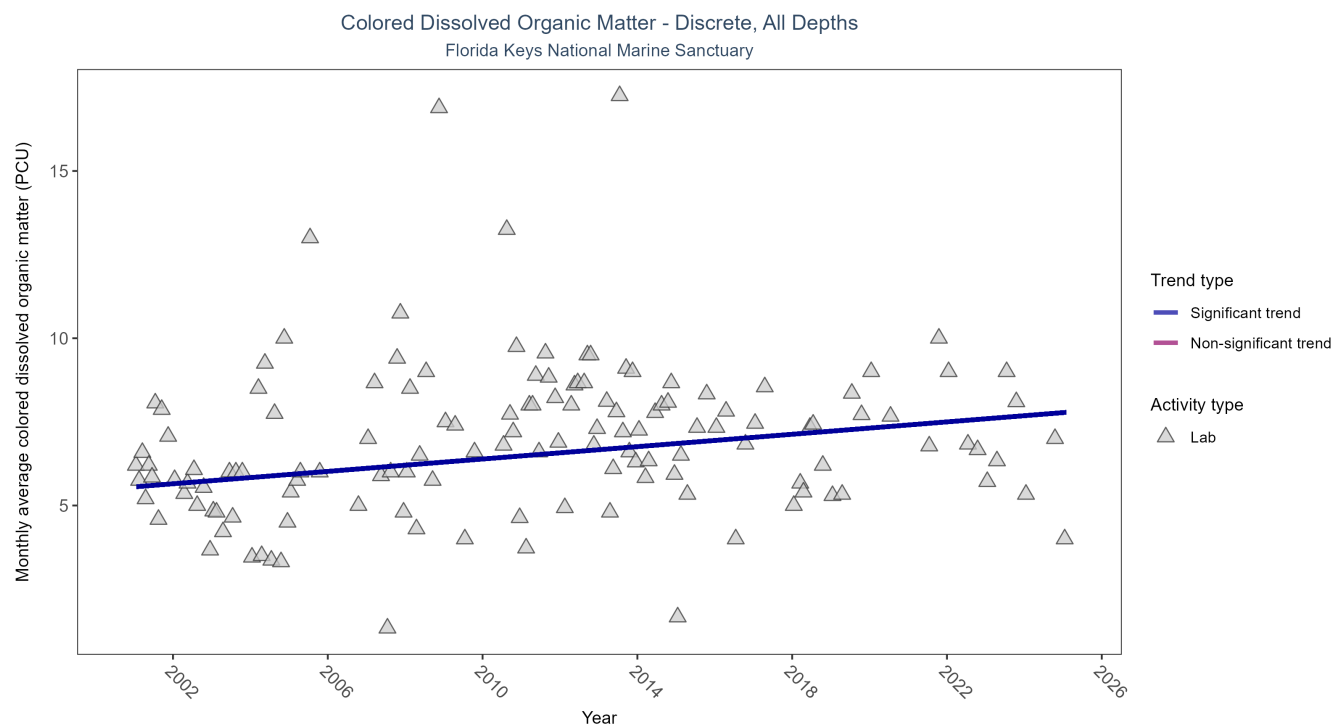


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1040	25	2001 - 2025	6	0.2251	5.5569	0.0925	0.0011

Monthly average colored dissolved organic matter increased by 0.09 PCU per year, indicating a decrease in water clarity.

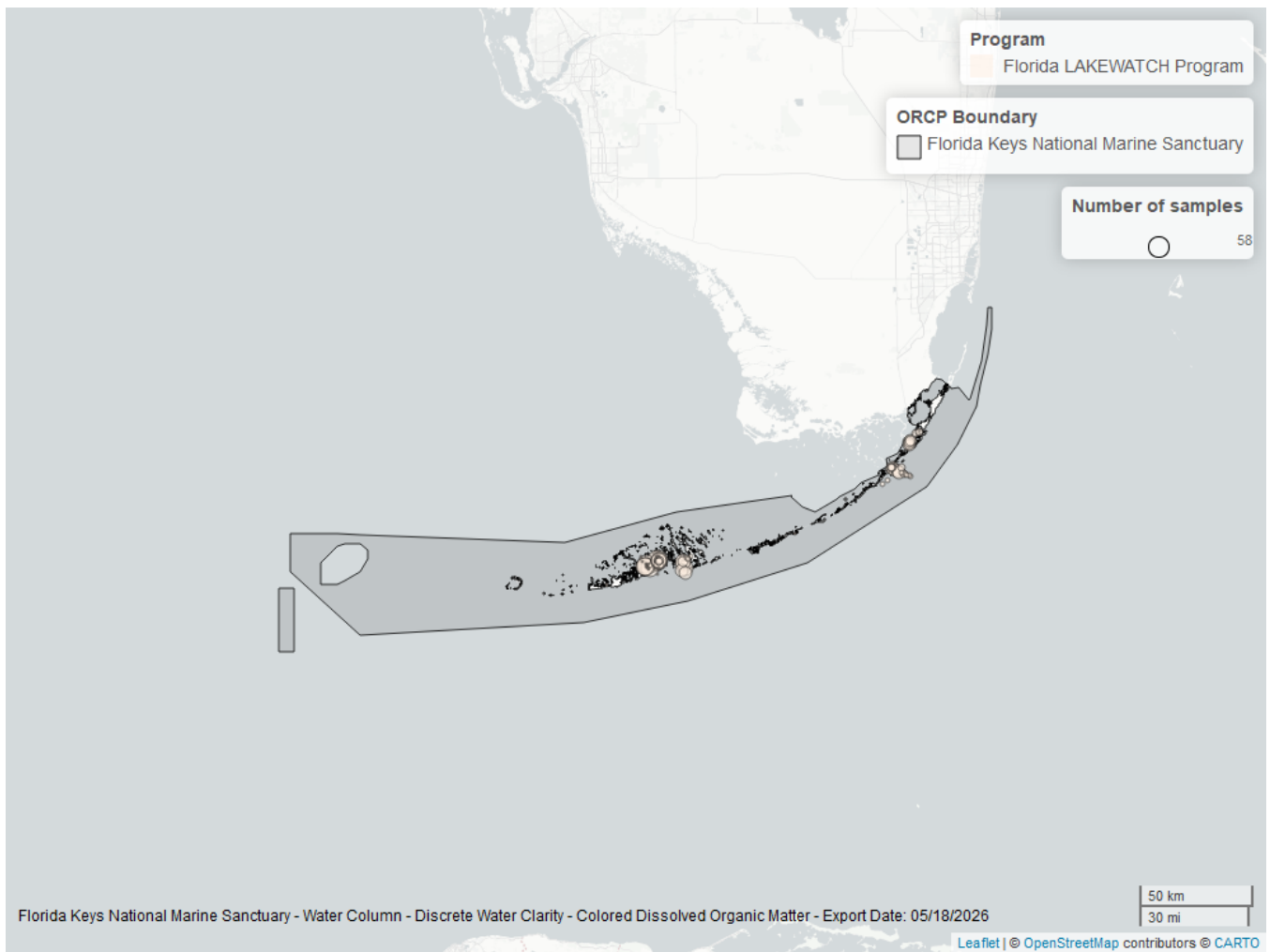


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
514	1040	2001	2025

Program names:

514 - Florida LAKEWATCH Program²

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

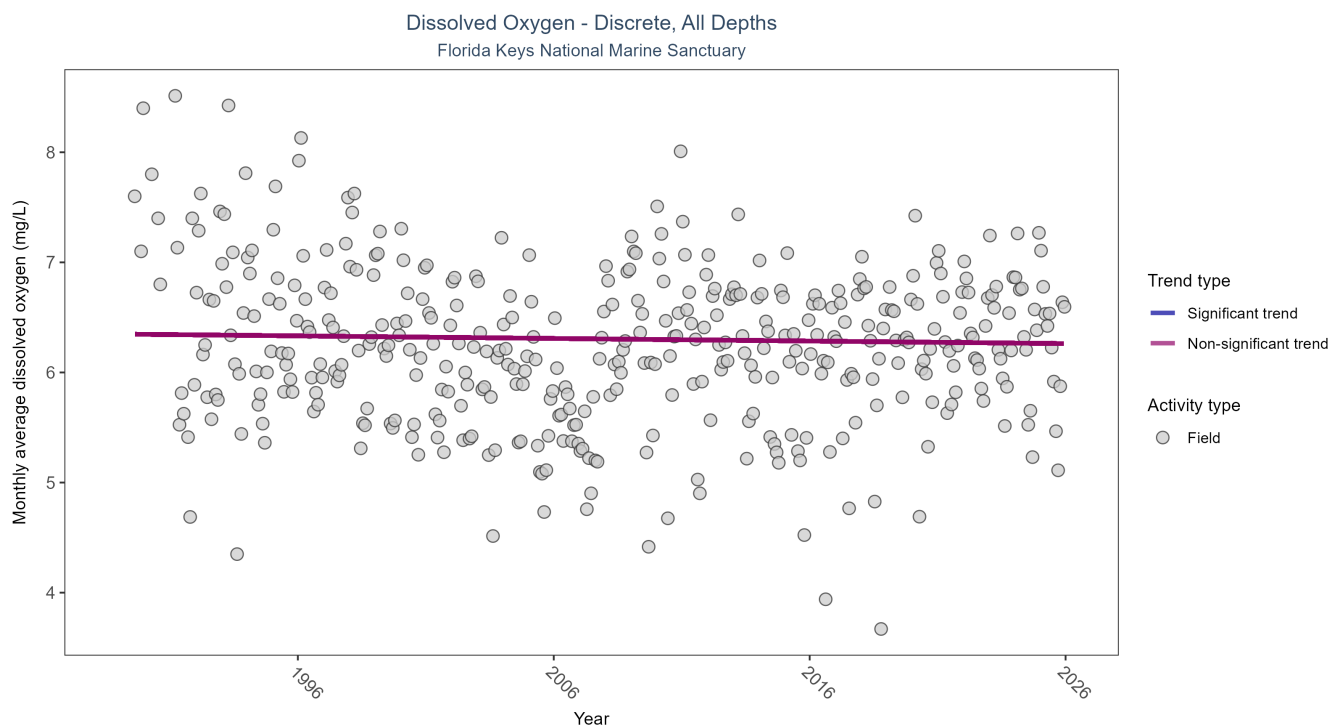


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	54647	37	1989 - 2025	6.3	-0.032	6.3486	-0.0023	0.3437

Dissolved oxygen showed no detectable trend between 1989 and 2025.

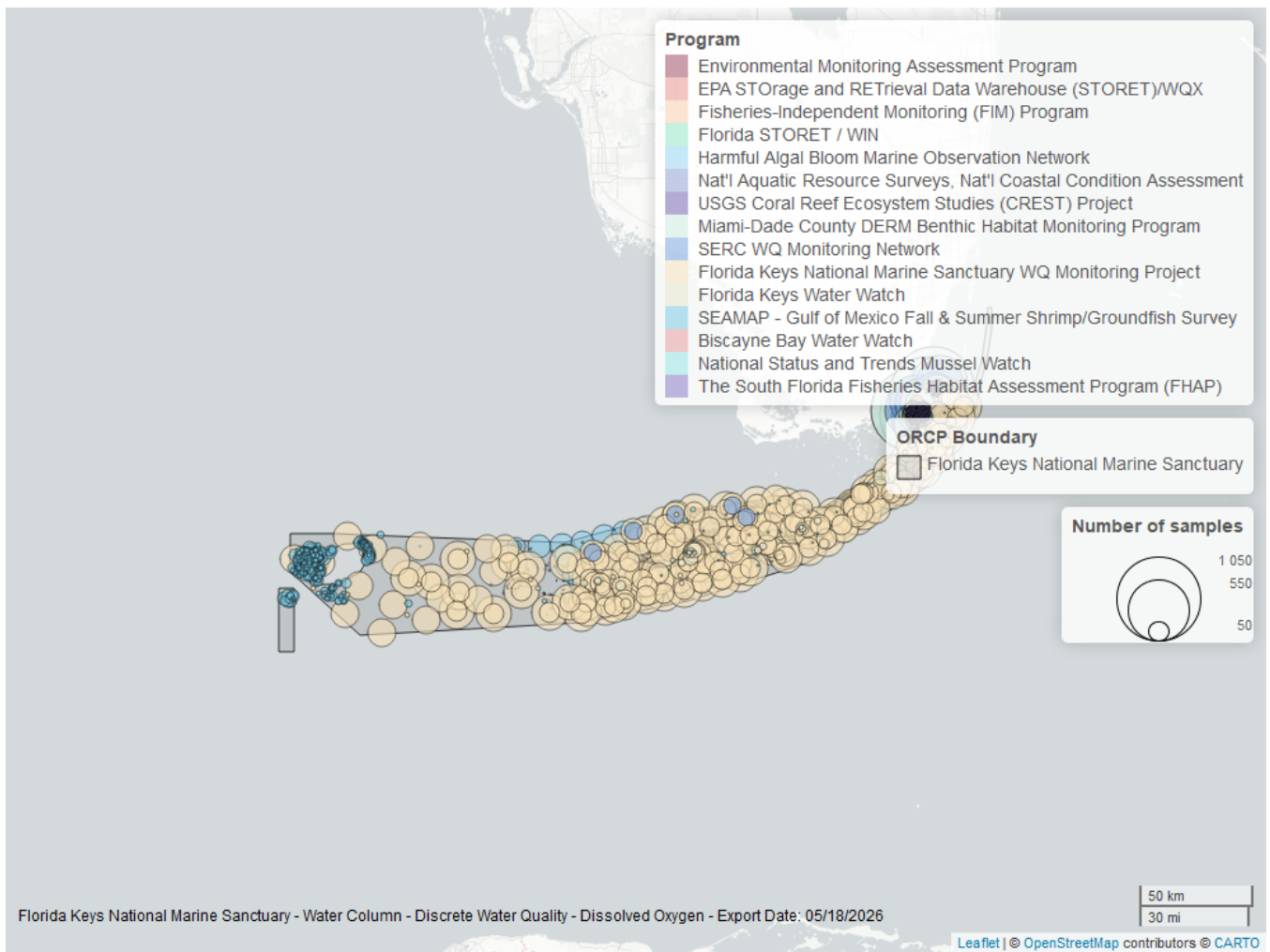


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
297	32263	1995	2023
4018	8832	1994	2025
5002	2966	2005	2025
60	2844	1993	2024
509	2590	1989	2008
69	1770	1997	2024
95	1560	1994	2018
4049	1131	2006	2025
3000	412	2015	2018
118	104	2000	2021
899	93	2014	2015
115	89	2000	2004
4057	59	2015	2018
102	42	1996	2000
103	28	2015	2015

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴
4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³
5002 - Florida STORET / WIN¹
60 - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey⁹
509 - SERC Water Quality Monitoring Network⁵
69 - Fisheries-Independent Monitoring (FIM) Program¹⁰
95 - Harmful Algal Bloom Marine Observation Network¹¹
4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹²
3000 - Florida Keys Water Watch¹³
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷
899 - USGS Coral Reef Ecosystem Studies (CREST) Project¹⁴
115 - Environmental Monitoring Assessment Program⁸
4057 - Biscayne Bay Water Watch¹⁵
102 - National Status and Trends Mussel Watch¹⁶
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

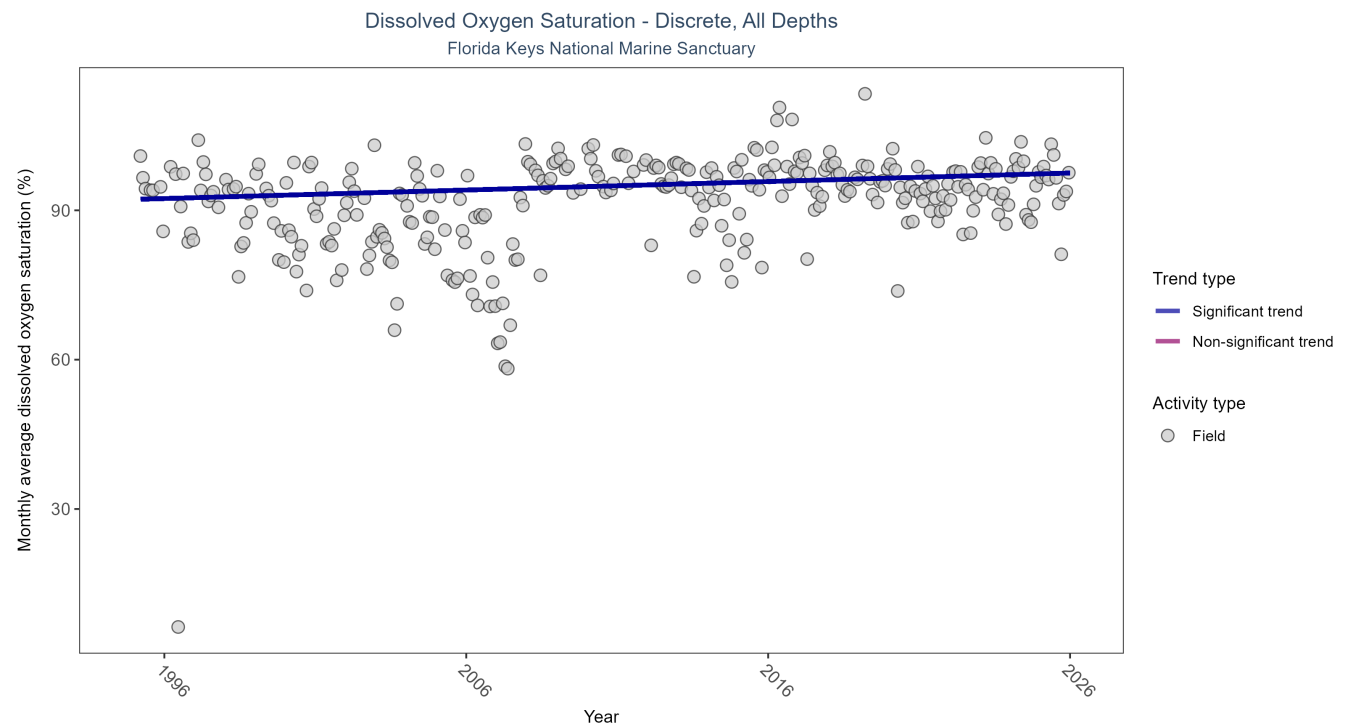


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	30525	31	1995 - 2025	94.7464	0.18	92.1918	0.171	0

Monthly average dissolved oxygen saturation increased by 0.17% per year.

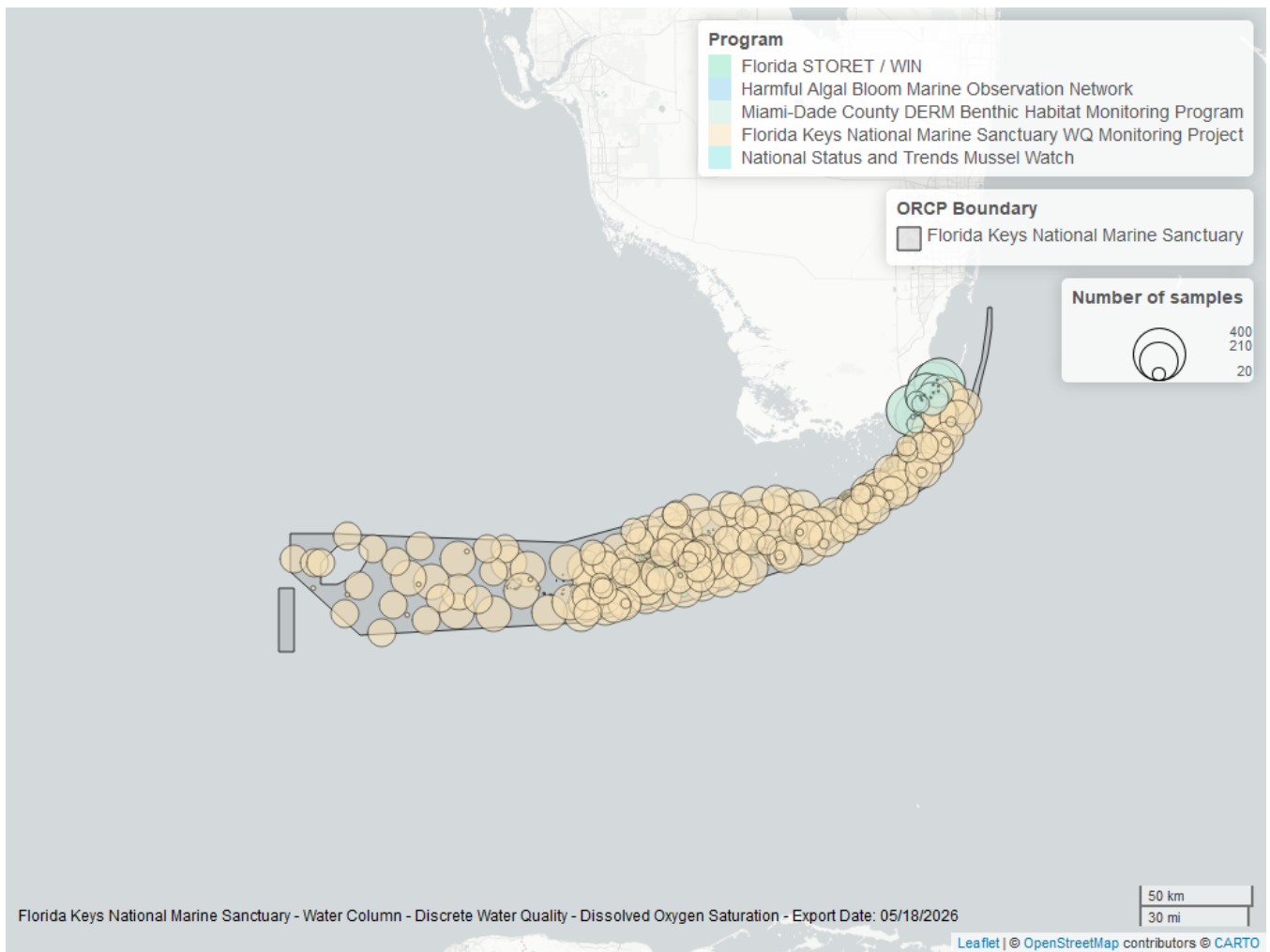


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
297	25475	1995	2020
5002	2683	2015	2025
4018	2411	2009	2025
102	18	1996	1996
95	1	2017	2017

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

5002 - Florida STORET / WIN¹

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

102 - National Status and Trends Mussel Watch¹⁶

95 - Harmful Algal Bloom Marine Observation Network¹¹

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

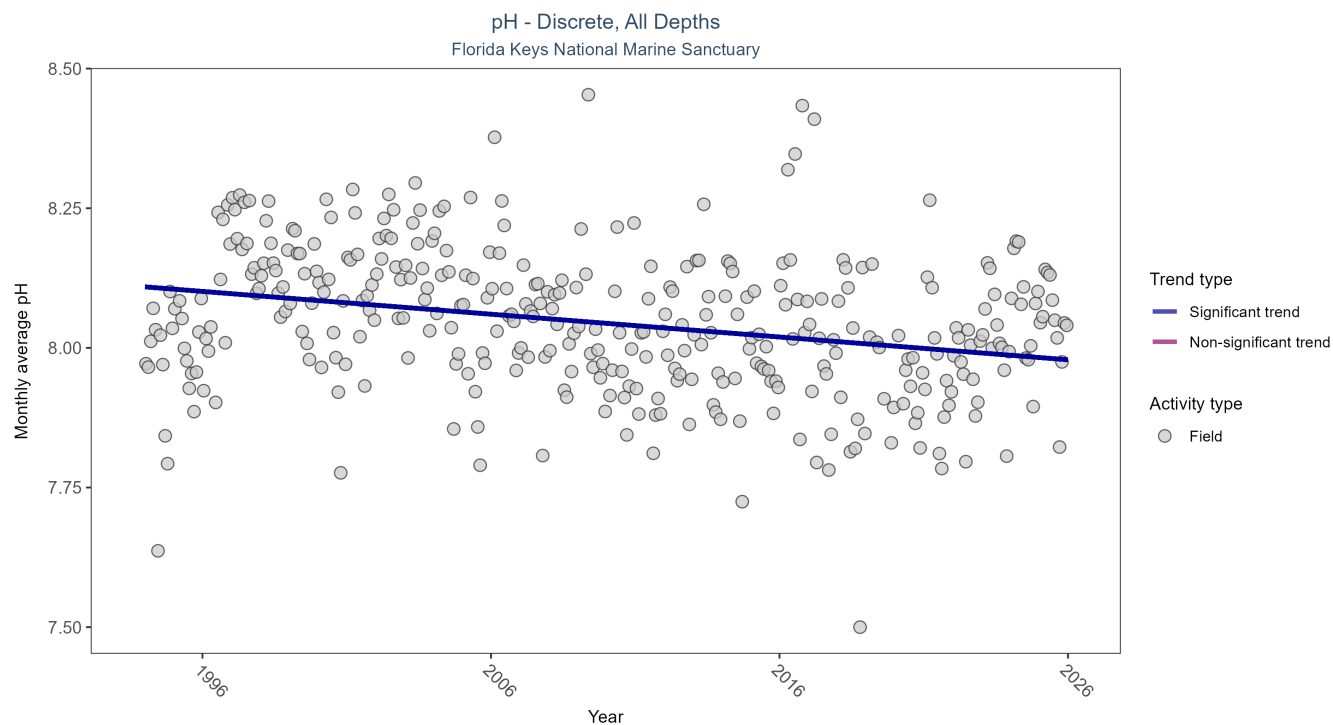


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	16699	32	1994 - 2025	8.08	-0.2158	8.1092	-0.0041	0

Monthly average pH decreased by less than 0.01 pH units per year.

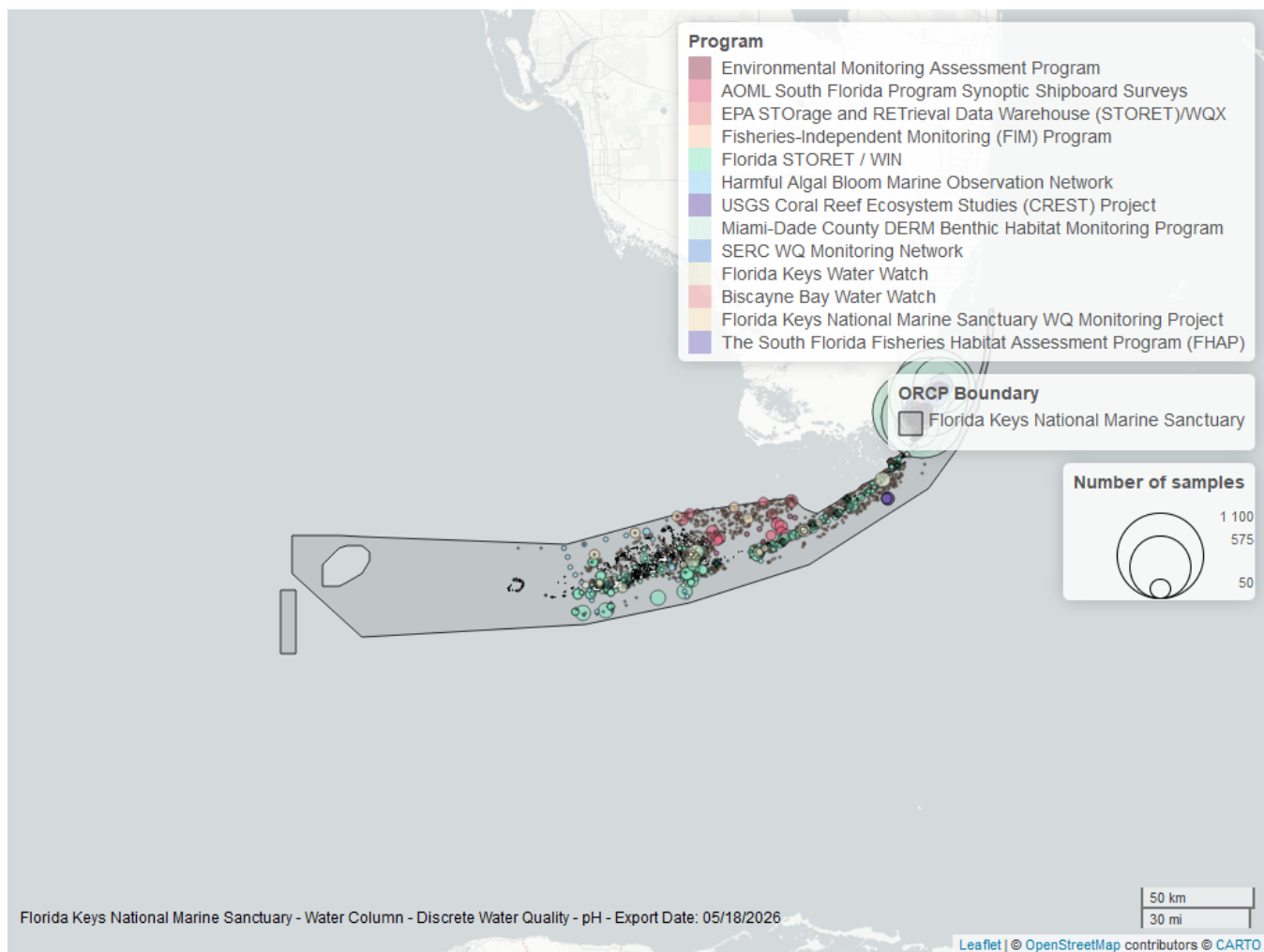


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
4018	9290	1994	2025
5002	2969	2005	2025
69	1760	1997	2024
4049	1210	2005	2025
509	467	2002	2008
3000	412	2015	2018
3	287	2009	2012
95	142	1994	2018
297	114	2003	2011
115	89	2000	2004
899	88	2014	2015
4057	59	2015	2018
103	30	2015	2015

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³
5002 - Florida STORET / WIN¹
69 - Fisheries-Independent Monitoring (FIM) Program¹⁰
4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹²
509 - SERC Water Quality Monitoring Network⁵
3000 - Florida Keys Water Watch¹³
3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹⁷
95 - Harmful Algal Bloom Marine Observation Network¹¹
297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴
115 - Environmental Monitoring Assessment Program⁸
899 - USGS Coral Reef Ecosystem Studies (CREST) Project¹⁴
4057 - Biscayne Bay Water Watch¹⁵
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

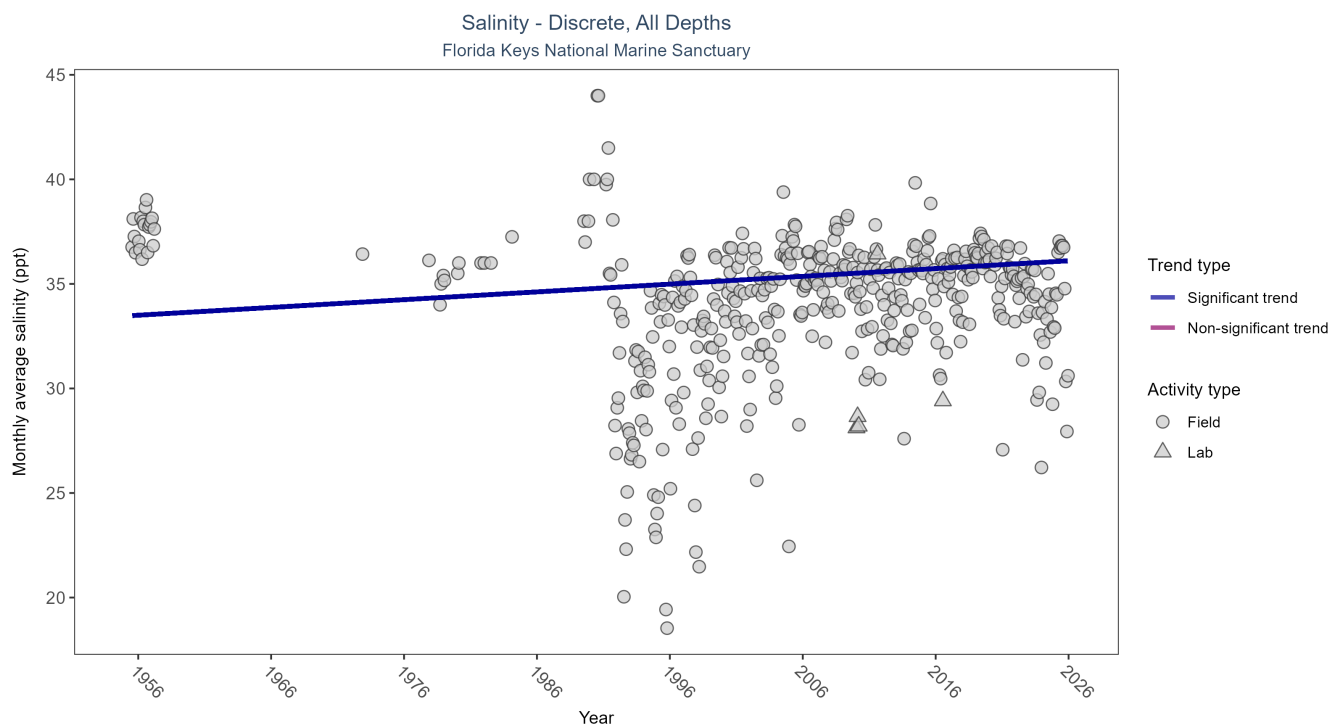


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	61634	48	1955 - 2025	36.0983	0.1064	33.4645	0.0372	0.0014

Monthly average salinity increased by 0.04 ppt per year.

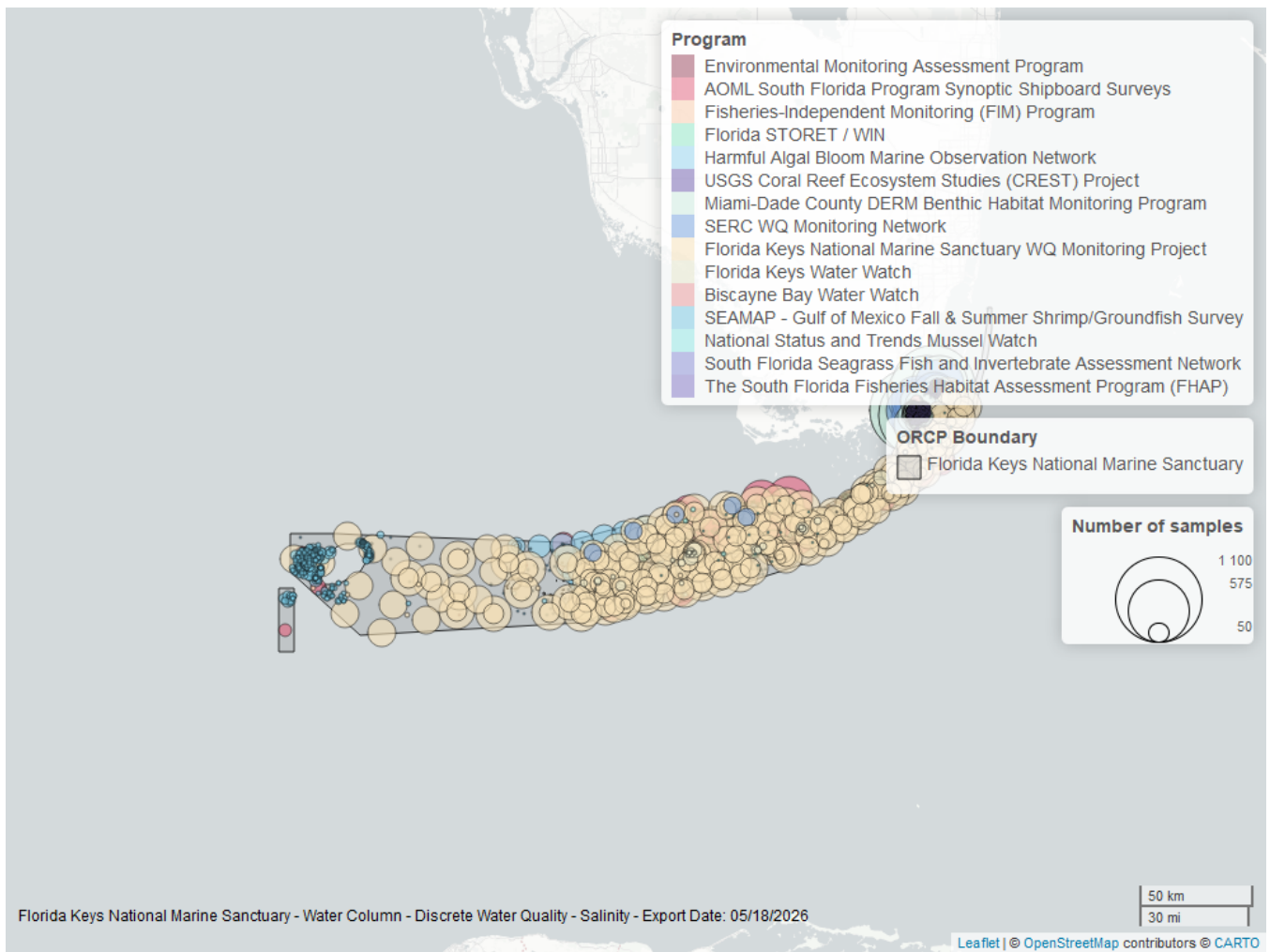


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
297	32002	1995	2023
4018	9400	1994	2025
3	5196	1998	2025
5002	2710	2005	2025
509	2581	1989	2008
965	2317	2005	2011
95	1888	1955	2018
60	1831	1993	2024
69	1768	1997	2024
4049	1275	2005	2025
3000	414	2015	2018
115	89	2000	2004
899	82	2014	2015
102	60	1996	2000
4057	59	2015	2018

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹⁷

5002 - Florida STORET / WIN¹

509 - SERC Water Quality Monitoring Network⁵

965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹⁸

95 - Harmful Algal Bloom Marine Observation Network¹¹

60 - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey⁹

69 - Fisheries-Independent Monitoring (FIM) Program¹⁰

4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹²

3000 - Florida Keys Water Watch¹³

115 - Environmental Monitoring Assessment Program⁸

899 - USGS Coral Reef Ecosystem Studies (CREST) Project¹⁴

102 - National Status and Trends Mussel Watch¹⁶

4057 - Biscayne Bay Water Watch¹⁵

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

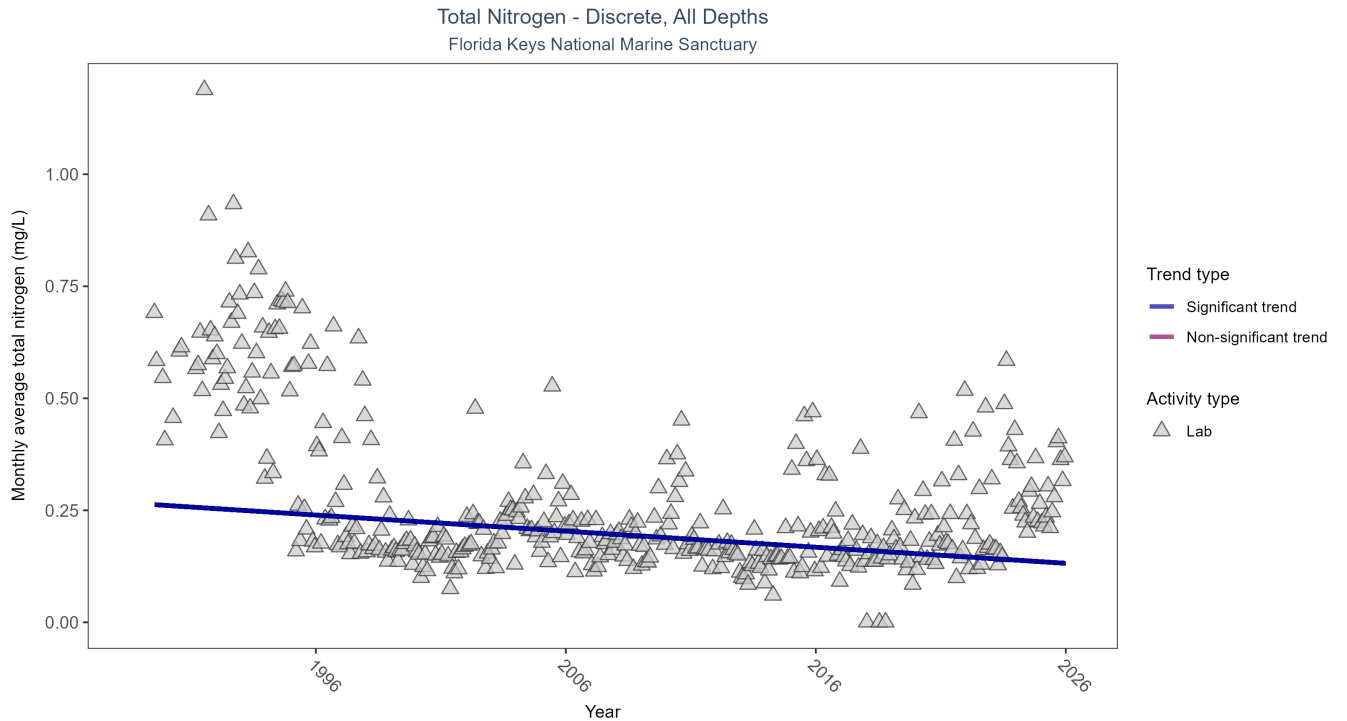


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	37584	37	1989 - 2025	0.1477	-0.2192	0.2646	-0.0036	0

Monthly average total nitrogen decreased by less than 0.01 mg/L per year.

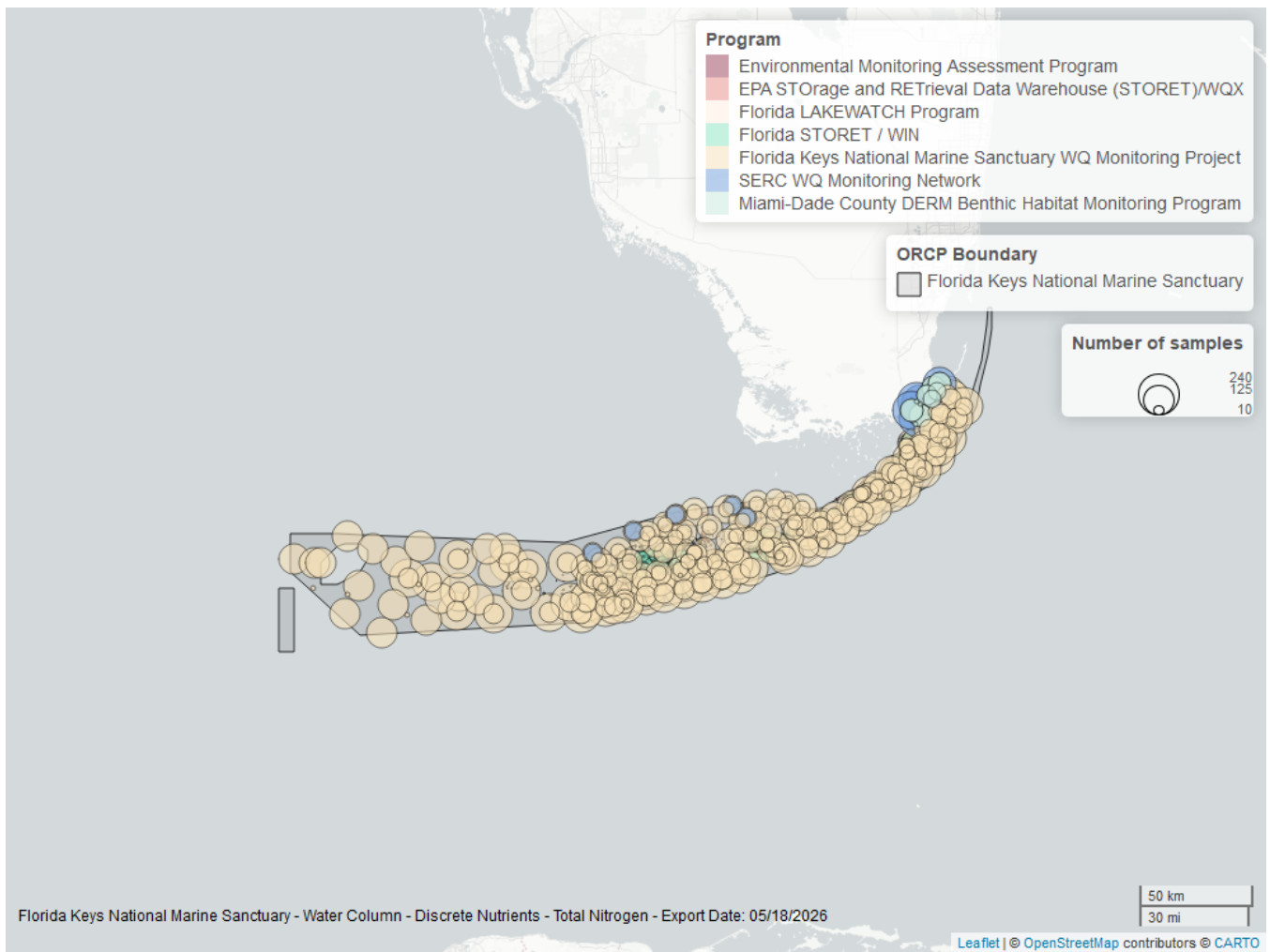


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
297	28759	1995	2023
5002	4692	1998	2025
514	2967	1998	2025
509	1424	1989	2008
4018	491	2009	2025
103	110	2000	2015
115	28	2000	2004

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

509 - SERC Water Quality Monitoring Network⁵

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

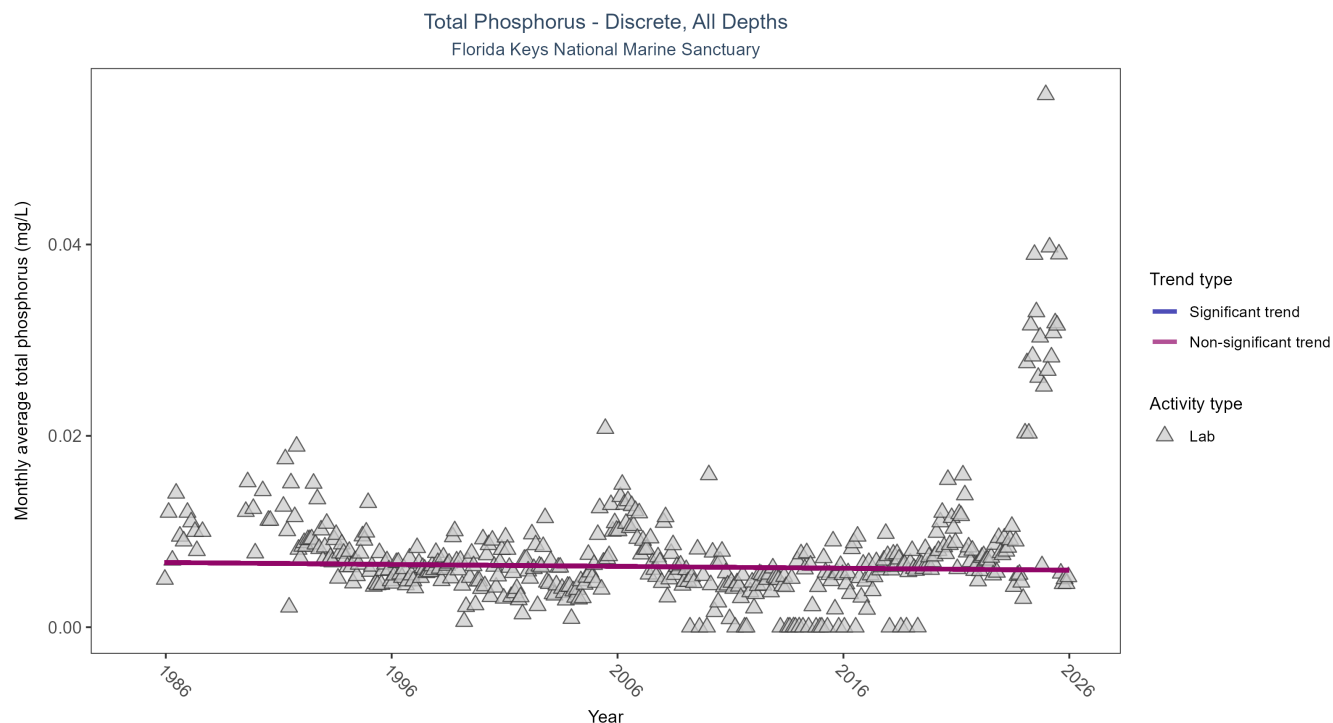


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	33102	40	1985 - 2025	0.006	-0.0362	0.0068	0	0.2865

Total phosphorus showed no detectable trend between 1985 and 2025.

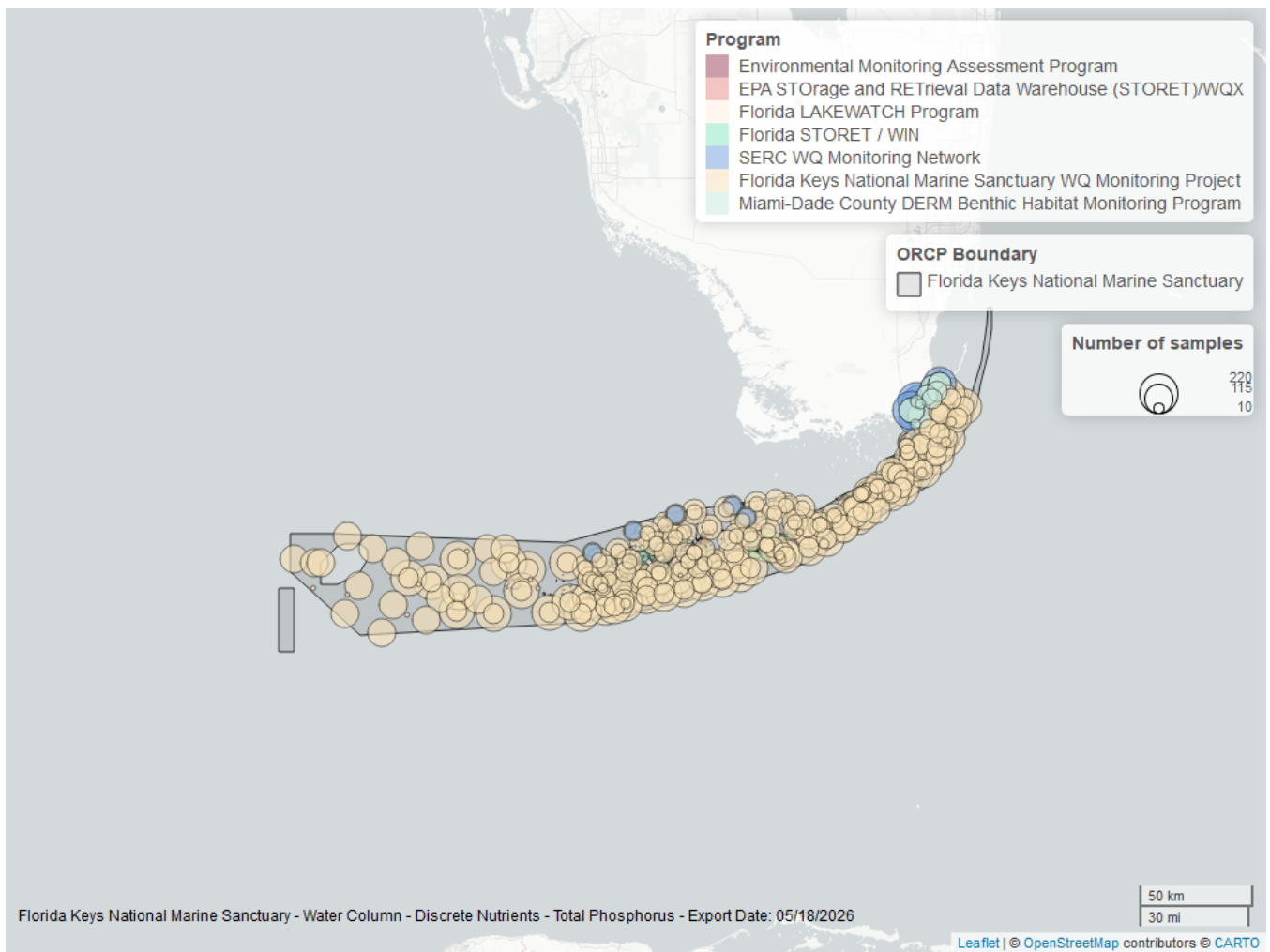


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
297	26166	1995	2023
514	2985	1998	2025
5002	2345	1985	2025
509	1425	1989	2008
4018	629	2004	2025
103	88	2000	2015
115	28	2000	2004

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

514 - Florida LAKEWATCH Program²

5002 - Florida STORET / WIN¹

509 - SERC Water Quality Monitoring Network⁵

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

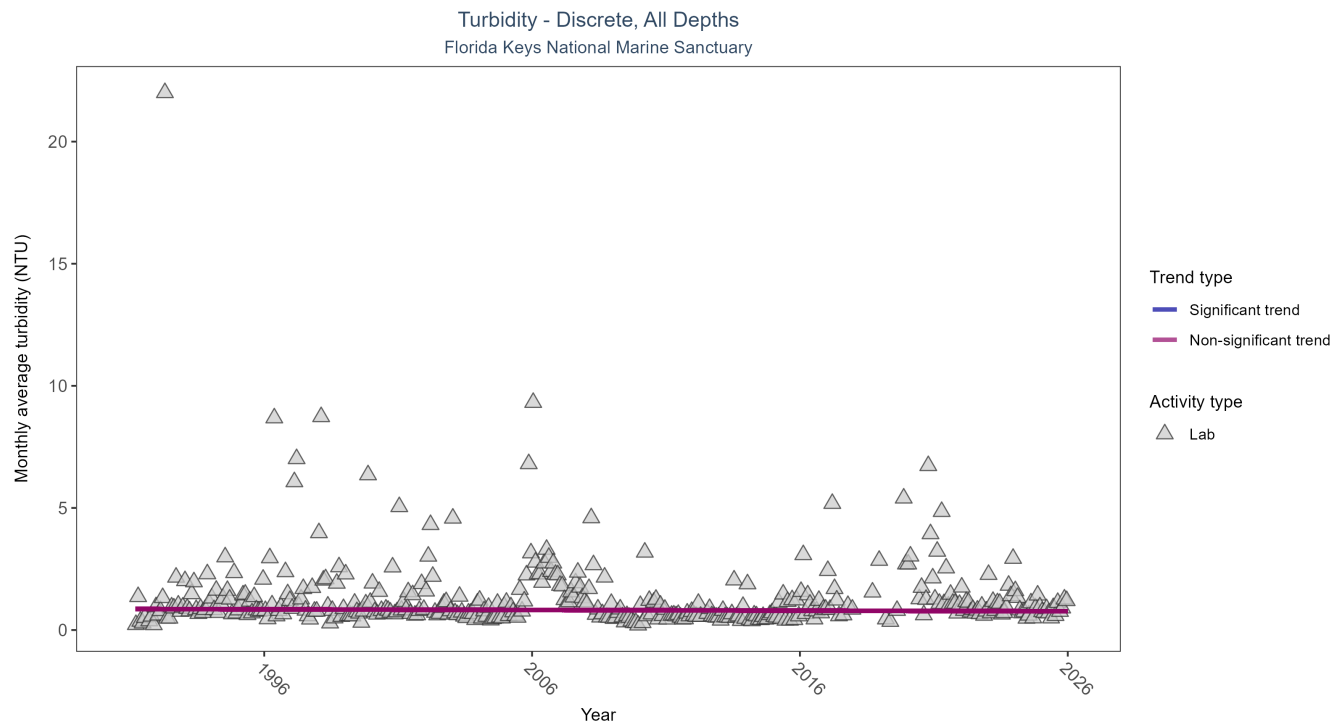


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3926	35	1991 - 2025	0.7	-0.0382	0.8615	-0.0027	0.2972

Turbidity showed no detectable trend between 1991 and 2025.

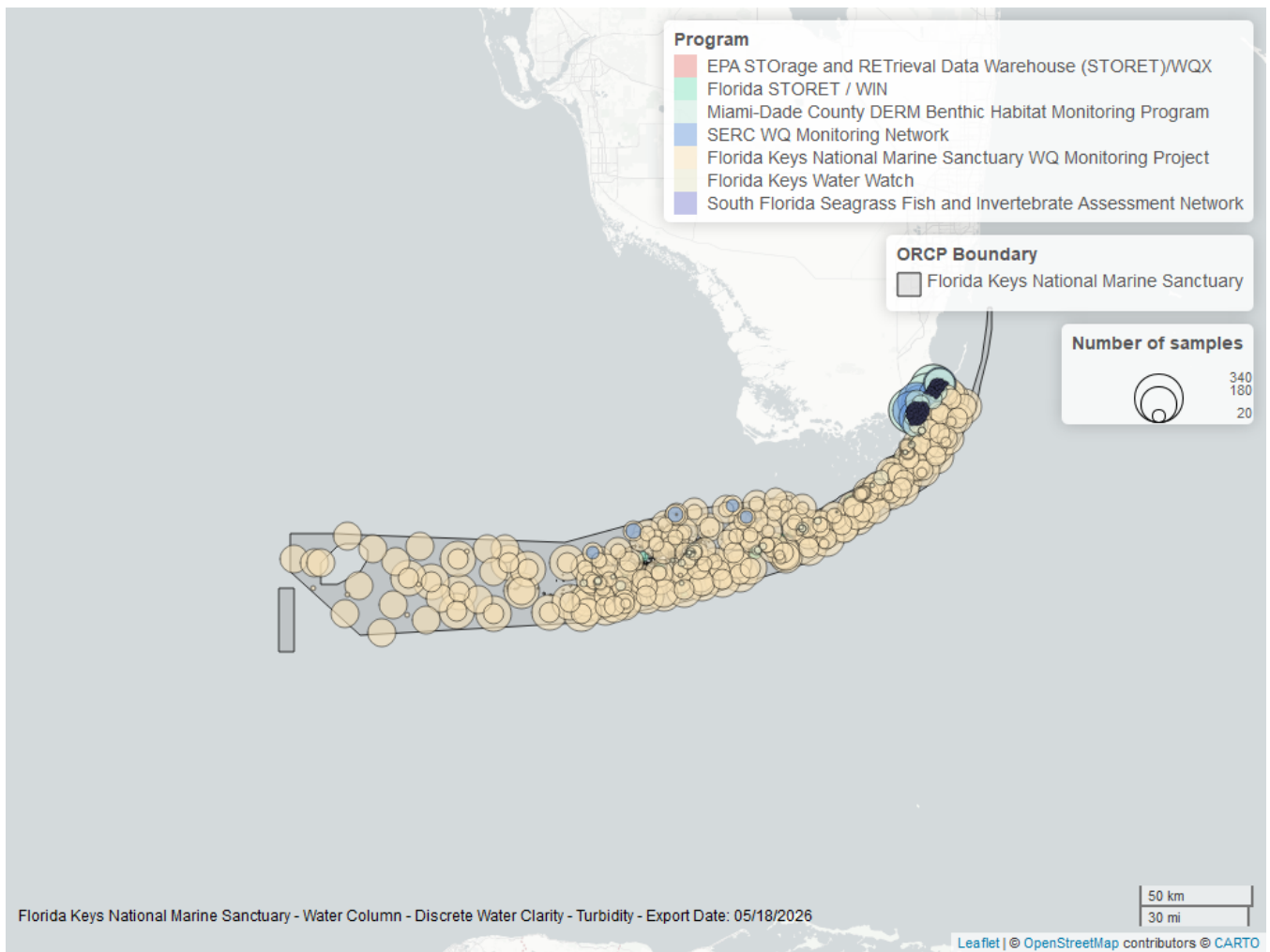


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
297	26746	1995	2023
4018	1813	1994	2025
509	1324	1991	2008
965	1157	2005	2011
5002	820	2010	2025
3000	405	2015	2018
103	17	2005	2006

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

509 - SERC Water Quality Monitoring Network⁵

965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹⁸

5002 - Florida STORET / WIN¹

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

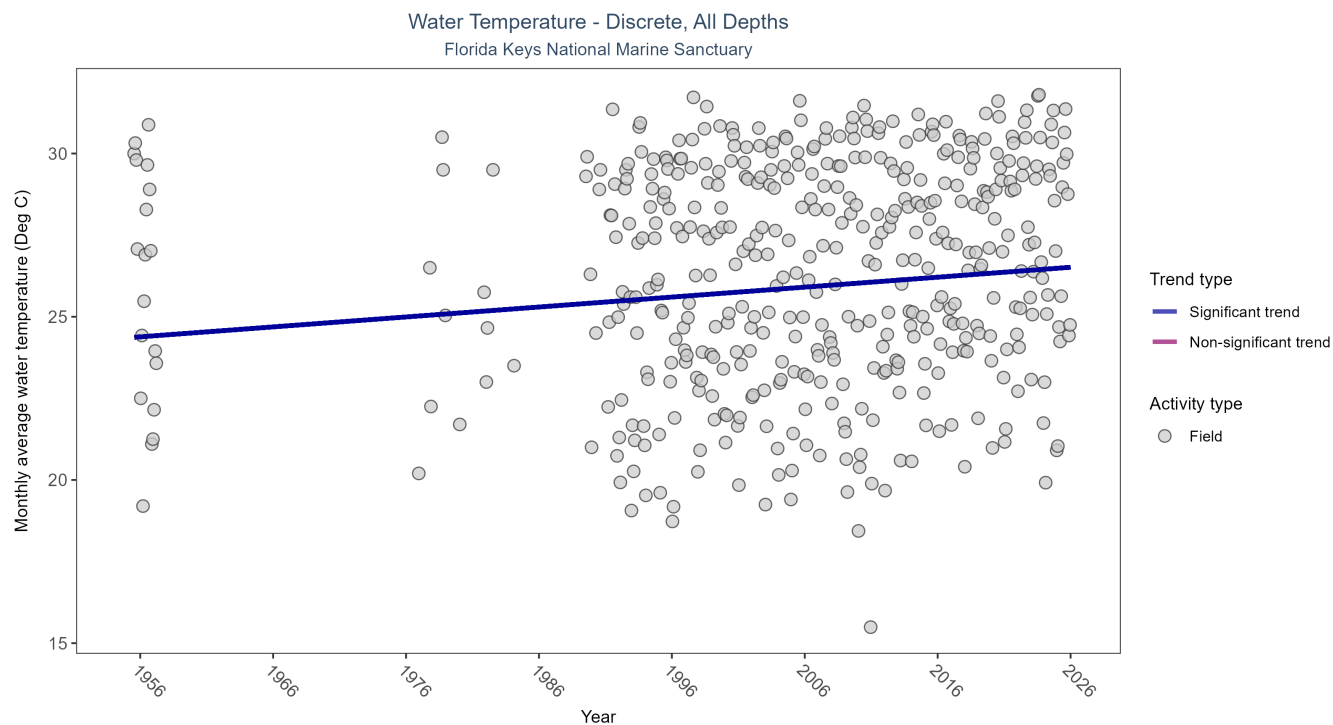


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	63298	47	1955 - 2025	27.2066	0.252	24.3534	0.0305	0

Monthly average water temperature increased by 0.03°C per year.

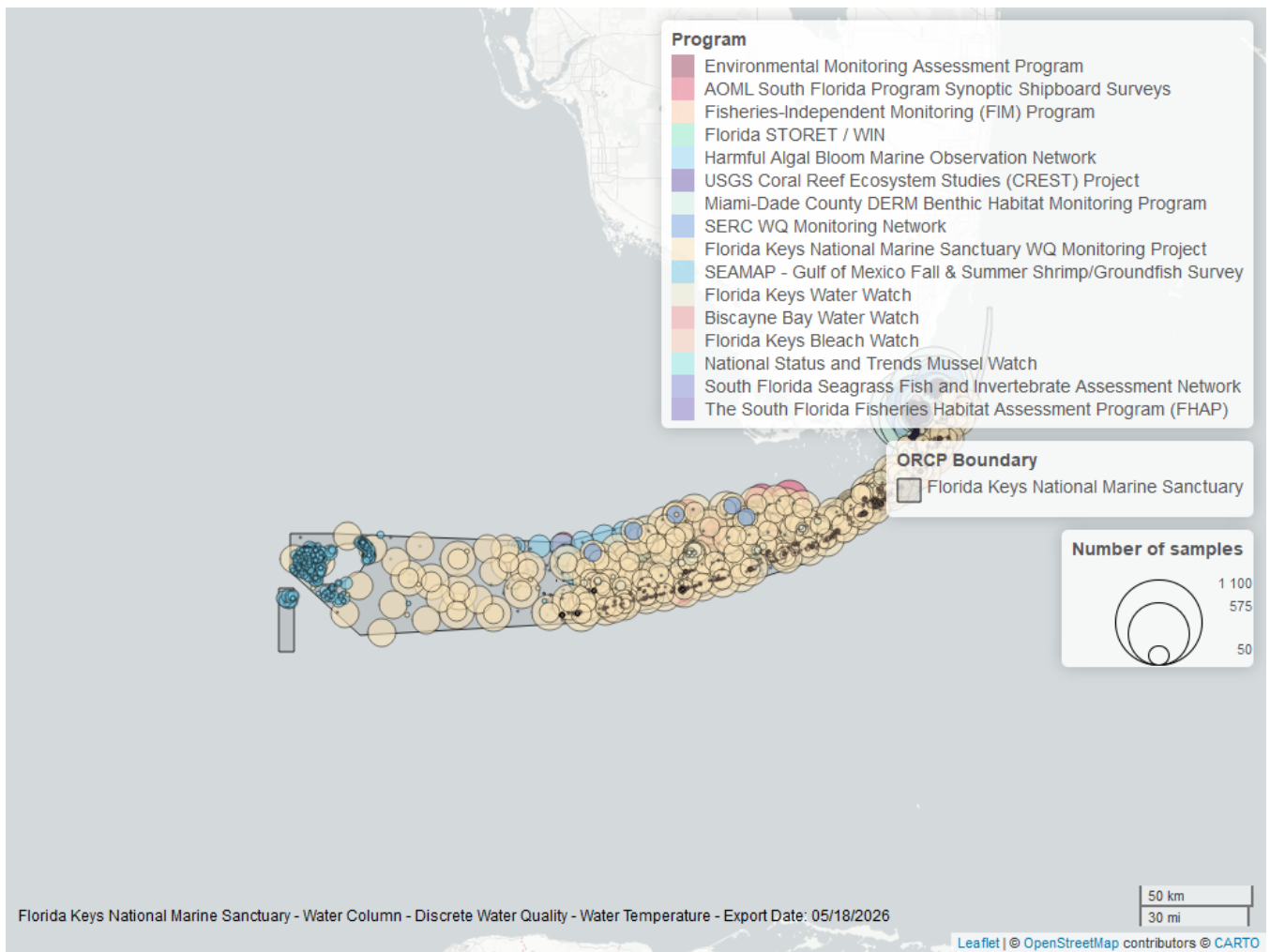


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
297	31986	1995	2023
4018	9362	1994	2025
3	3630	1998	2025
60	3306	1993	2024
5002	3234	2005	2025
509	2591	1989	2008
965	2317	2005	2011
95	1956	1955	2018
69	1803	1997	2024
4049	1275	2005	2025
982	1200	2014	2024
3000	409	2015	2018
115	89	2000	2004
899	85	2014	2015
4057	59	2015	2018

ProgramID	N_Data	YearMin	YearMax
102	43	1996	2000

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project⁴

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹⁷

60 - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey⁹

5002 - Florida STORET / WIN¹

509 - SERC Water Quality Monitoring Network⁵

965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹⁸

95 - Harmful Algal Bloom Marine Observation Network¹¹

69 - Fisheries-Independent Monitoring (FIM) Program¹⁰

4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹²

982 - Florida Keys Bleach Watch¹⁹

3000 - Florida Keys Water Watch¹³

115 - Environmental Monitoring Assessment Program⁸

899 - USGS Coral Reef Ecosystem Studies (CREST) Project¹⁴

4057 - Biscayne Bay Water Watch¹⁵

102 - National Status and Trends Mussel Watch¹⁶

Water Quality - Continuous

The following files were used in the continuous analysis:

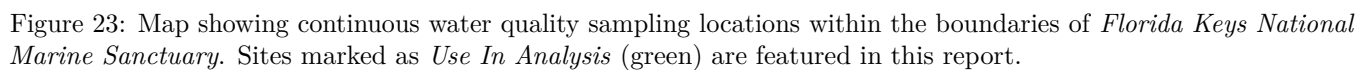
- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Florida Keys National Marine Sanctuary

Table 28: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
2	1B	6	TRUE	Water Temperature , Salinity , Specific Conductivity
5	KYWF1	21	TRUE	Water Temperature
5	LONF1	28	TRUE	Water Temperature
5	MLRF1	33	TRUE	Water Temperature
5	SANF1	15	TRUE	Water Temperature
5	SMKF1	21	TRUE	Water Temperature
7	245323080410100	3	FALSE	Water Temperature, Salinity
7	245622080364200	3	FALSE	Water Temperature, Salinity
296	214	18	TRUE	Water Temperature
296	215	16	TRUE	Water Temperature
296	216	17	TRUE	Water Temperature
296	220	17	TRUE	Water Temperature
296	223	18	TRUE	Water Temperature
296	225	18	TRUE	Water Temperature
296	227	17	TRUE	Water Temperature
296	235	18	TRUE	Water Temperature
296	237	18	TRUE	Water Temperature
296	239	17	TRUE	Water Temperature
296	241	18	TRUE	Water Temperature
296	243	18	TRUE	Water Temperature
296	248	18	TRUE	Water Temperature
296	255	18	TRUE	Water Temperature
296	260	18	TRUE	Water Temperature
296	267	18	TRUE	Water Temperature
296	269	18	TRUE	Water Temperature
296	271	18	TRUE	Water Temperature
296	273	18	TRUE	Water Temperature
296	276	18	TRUE	Water Temperature
296	284	18	TRUE	Water Temperature
296	285	18	TRUE	Water Temperature
296	287	18	TRUE	Water Temperature
296	291	18	TRUE	Water Temperature
296	294	18	TRUE	Water Temperature
296	296	18	TRUE	Water Temperature
296	305	18	TRUE	Water Temperature
296	307	18	TRUE	Water Temperature
296	309	18	TRUE	Water Temperature
296	314	18	TRUE	Water Temperature
296	500	8	TRUE	Water Temperature
296	501	7	TRUE	Water Temperature
296	502	4	FALSE	Water Temperature
296	503	1	FALSE	Water Temperature
296	504	1	FALSE	Water Temperature
296	506	8	TRUE	Water Temperature
296	507	8	TRUE	Water Temperature
296	508	8	TRUE	Water Temperature
296	509	8	TRUE	Water Temperature
296	SB	19	TRUE	Water Temperature
899	Crocker	10	TRUE	Water Temperature
899	Molasses	5	TRUE	Water Temperature
899	Sombrero	14	TRUE	Water Temperature
986	10	5	TRUE	Water Temperature
986	11	22	TRUE	Water Temperature
986	12	17	TRUE	Water Temperature
986	14	23	TRUE	Water Temperature
986	15	19	TRUE	Water Temperature
986	18	9	TRUE	Water Temperature
986	21	9	TRUE	Water Temperature
986	22	16	TRUE	Water Temperature
986	23	13	TRUE	Water Temperature
986	24	15	TRUE	Water Temperature
986	25	15	TRUE	Water Temperature
986	26	16	TRUE	Water Temperature
986	30	13	TRUE	Water Temperature
986	32	22	TRUE	Water Temperature
986	33	9	TRUE	Water Temperature
986	34	23	TRUE	Water Temperature
986	35	19	TRUE	Water Temperature
986	36	18	TRUE	Water Temperature
986	37	9	TRUE	Water Temperature
986	38	23	TRUE	Water Temperature
986	39	7	TRUE	Water Temperature
986	40	23	TRUE	Water Temperature
986	50	12	TRUE	Water Temperature
986	51	22	TRUE	Water Temperature
986	52	17	TRUE	Water Temperature
986	53	17	TRUE	Water Temperature
986	54	13	TRUE	Water Temperature
986	55	23	TRUE	Water Temperature
986	56	19	TRUE	Water Temperature
986	57	17	TRUE	Water Temperature
986	58	11	TRUE	Water Temperature
986	59	23	TRUE	Water Temperature
986	60	16	TRUE	Water Temperature
986	61	9	TRUE	Water Temperature
986	70	12	TRUE	Water Temperature
986	72	16	TRUE	Water Temperature
986	73	17	TRUE	Water Temperature
986	74	13	TRUE	Water Temperature
986	75	15	TRUE	Water Temperature
986	76	16	TRUE	Water Temperature
986	77	17	TRUE	Water Temperature
986	78	11	TRUE	Water Temperature
986	79	18	TRUE	Water Temperature
986	80	16	TRUE	Water Temperature
986	81	9	TRUE	Water Temperature
986	83	18	TRUE	Water Temperature
989	FKNMS-200YR-HD	12	TRUE	Water Temperature
989	FKNMS-7MILE-BR	20	TRUE	Water Temperature
989	FKNMS-9FT-SHOAL	21	TRUE	Water Temperature
989	FKNMS-ALLIGATOR	21	TRUE	Water Temperature
989	FKNMS-BHONDA-BR	22	TRUE	Water Temperature

2 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Moored Instrument Array²⁰
5 - National Data Buoy Center²¹
7 - National Water Information System²²
296 - Florida Keys National Marine Sanctuary Seagrass Monitoring Project²³
899 - USGS Coral Reef Ecosystem Studies (CREST) Project¹⁴
986 - Water Temperature on Coral Reefs in the Florida Keys²⁴
989 - Continuous Bottom Temperature Measurements along the Florida Reef Tract²⁵
10004 - Florida Keys Aquatic Preserves Continuous Water Quality Monitoring²⁶



pH - Continuous

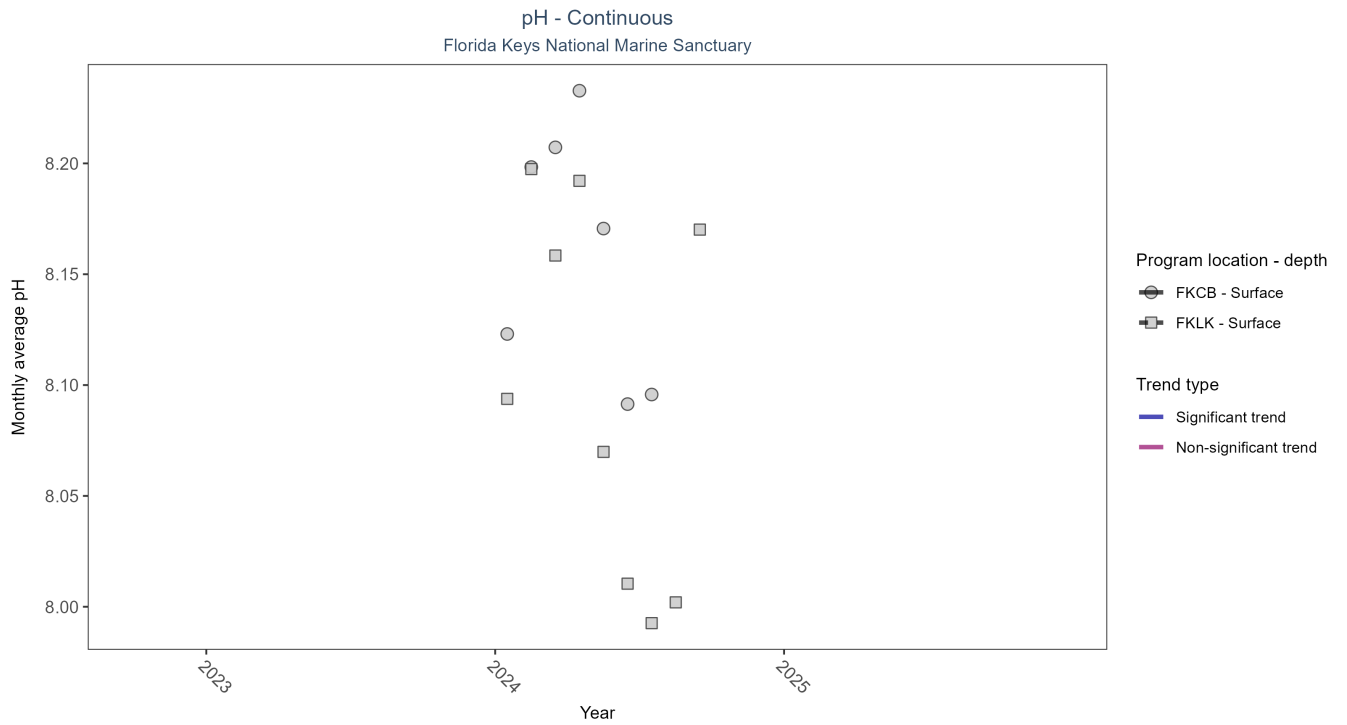


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16263	1	2024 - 2024	8.2	-	-	-	-
FKLK	Insufficient data to calculate trend	21517	1	2024 - 2024	8.1	-	-	-	-

There was insufficient data to fit a model for two locations.

Dissolved Oxygen Saturation - Continuous

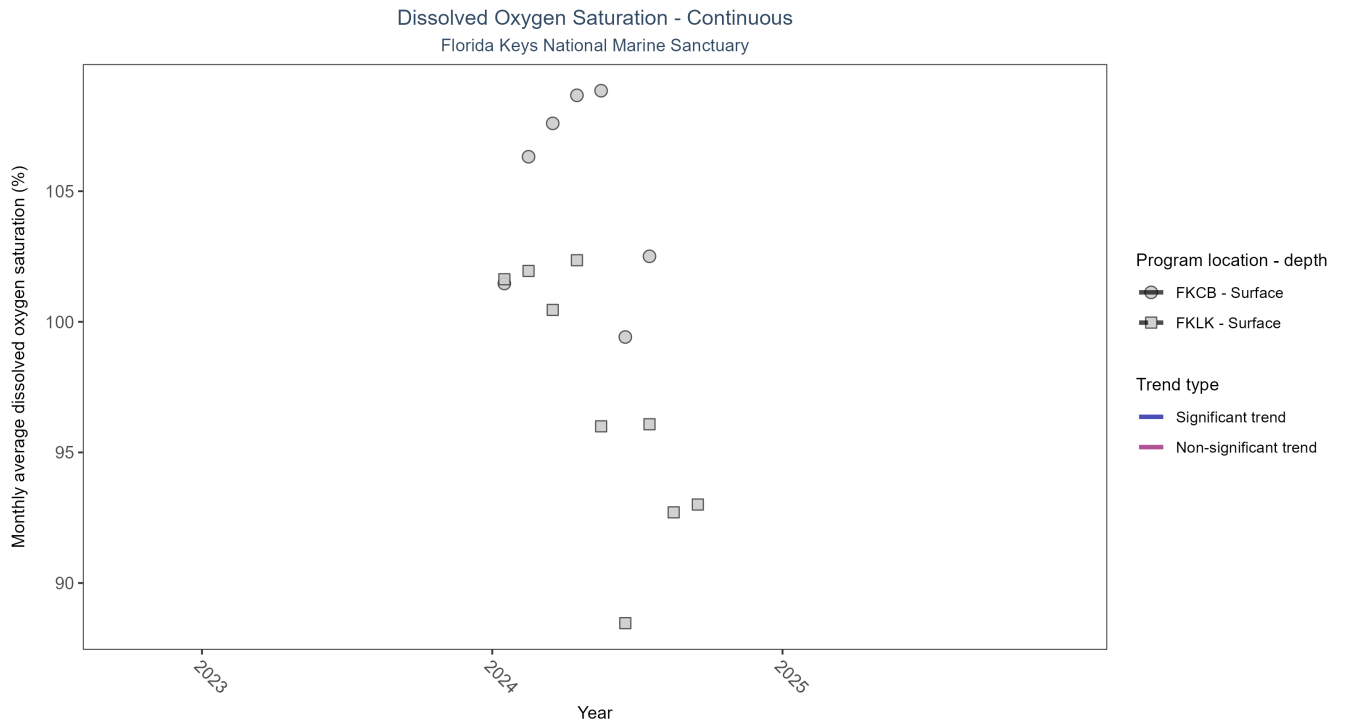


Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16263	1	2024 - 2024	103.3	-	-	-	-
FKLK	Insufficient data to calculate trend	21525	1	2024 - 2024	91.9	-	-	-	-

There was insufficient data to fit a model for two locations.

Dissolved Oxygen - Continuous

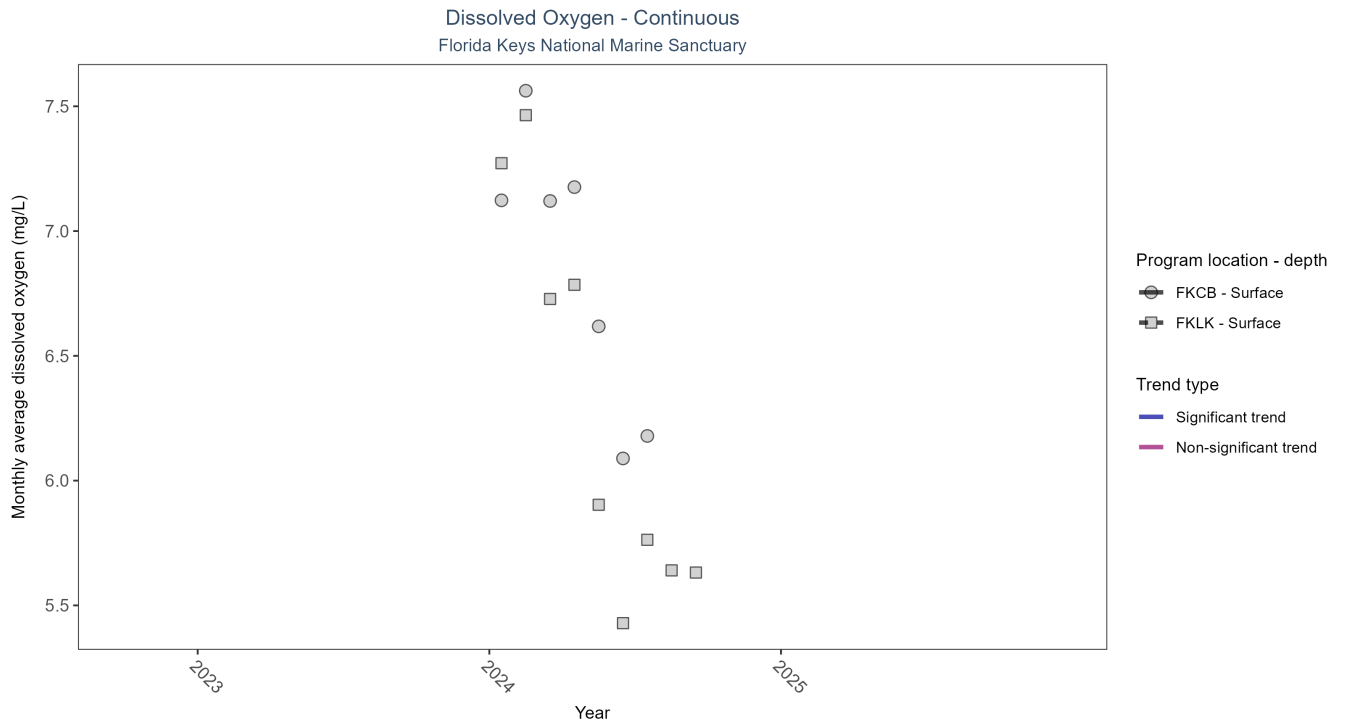


Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16262	1	2024 - 2024	6.8	-	-	-	-
FKLK	Insufficient data to calculate trend	21525	1	2024 - 2024	6.2	-	-	-	-

There was insufficient data to fit a model for two locations.

Turbidity - Continuous

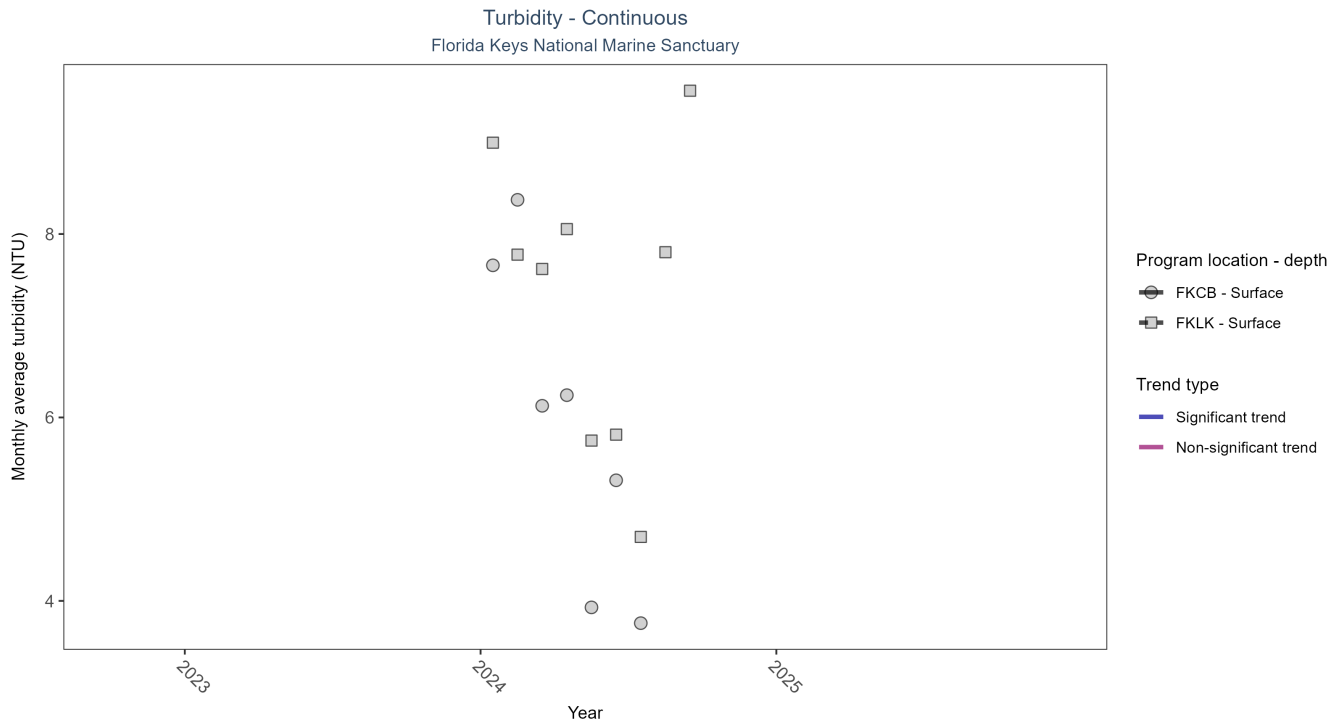


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16240	1	2024 - 2024	4	-	-	-	-
FKLK	Insufficient data to calculate trend	21399	1	2024 - 2024	6	-	-	-	-

There was insufficient data to fit a model for two locations.

Water Temperature - Continuous - Program 2

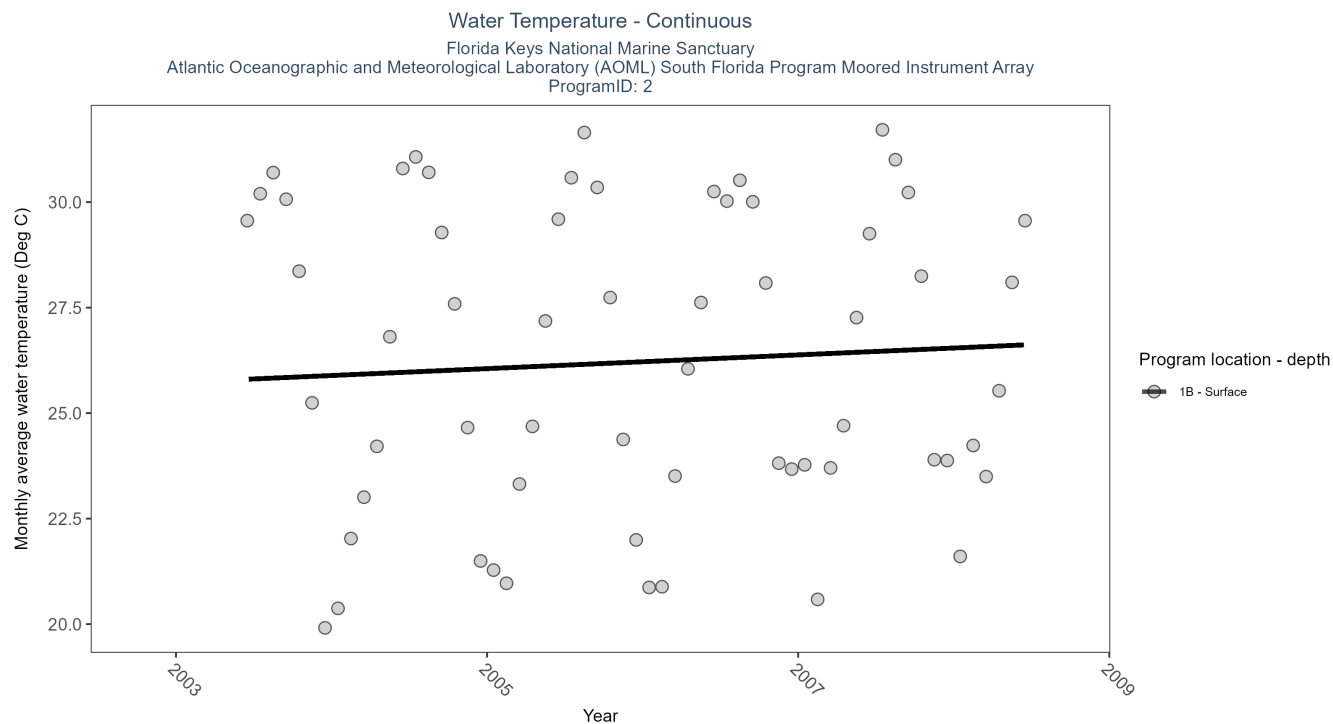


Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - Program 2

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
1B	Significantly increasing trend	86204	6	2003 - 2008	26.38	0.26	25.73	0.16	0.04

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

Water Temperature - Continuous - Program 5

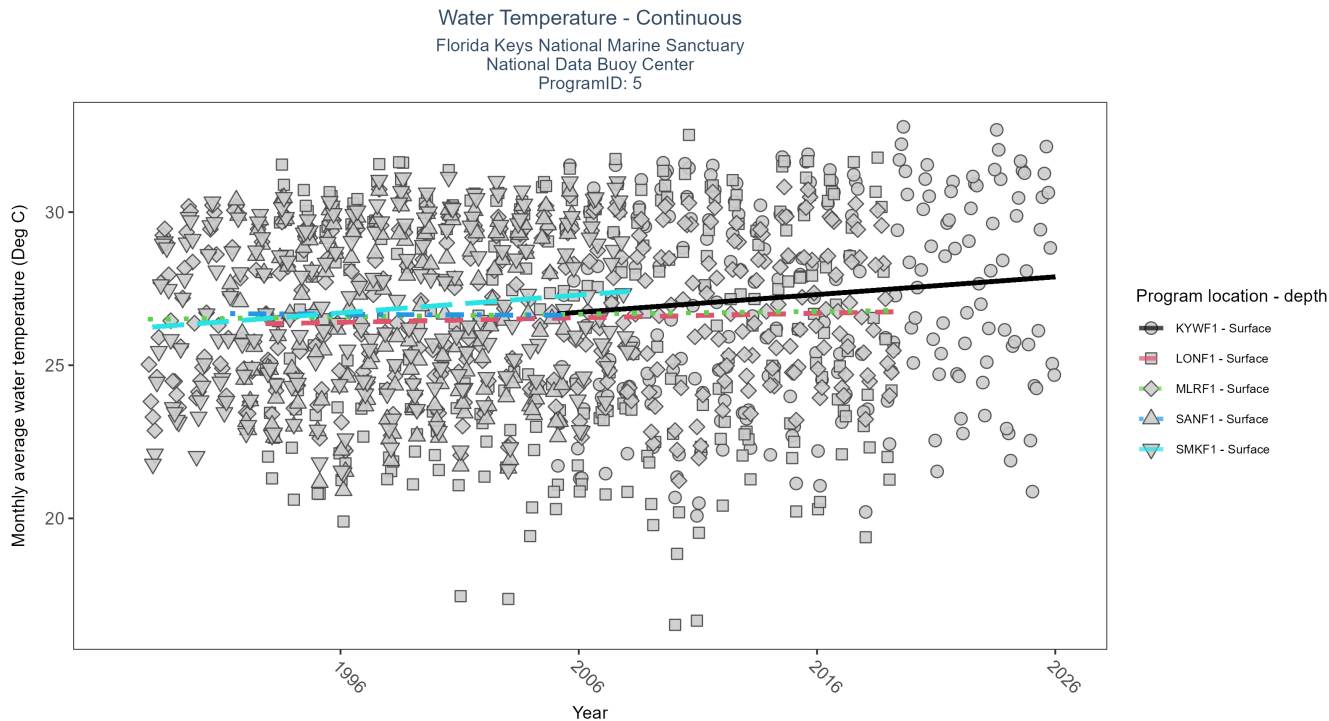


Figure 29: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Water Temperature - Program 5

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
KYWF1	Significantly increasing trend	1565245	21	2005 - 2025	27.5	0.29	26.67	0.06	0.00
LONF1	No significant trend	205971	28	1992 - 2019	26.6	0.07	26.34	0.01	0.08
MLRF1	Significantly increasing trend	256798	33	1987 - 2019	26.5	0.10	26.49	0.01	0.00
SANF1	No significant trend	117833	15	1991 - 2005	26.7	-0.03	26.69	0.00	0.62
SMKF1	Significantly increasing trend	154326	21	1988 - 2008	26.8	0.34	26.24	0.06	0.00

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

Water Temperature - Continuous - Program 7

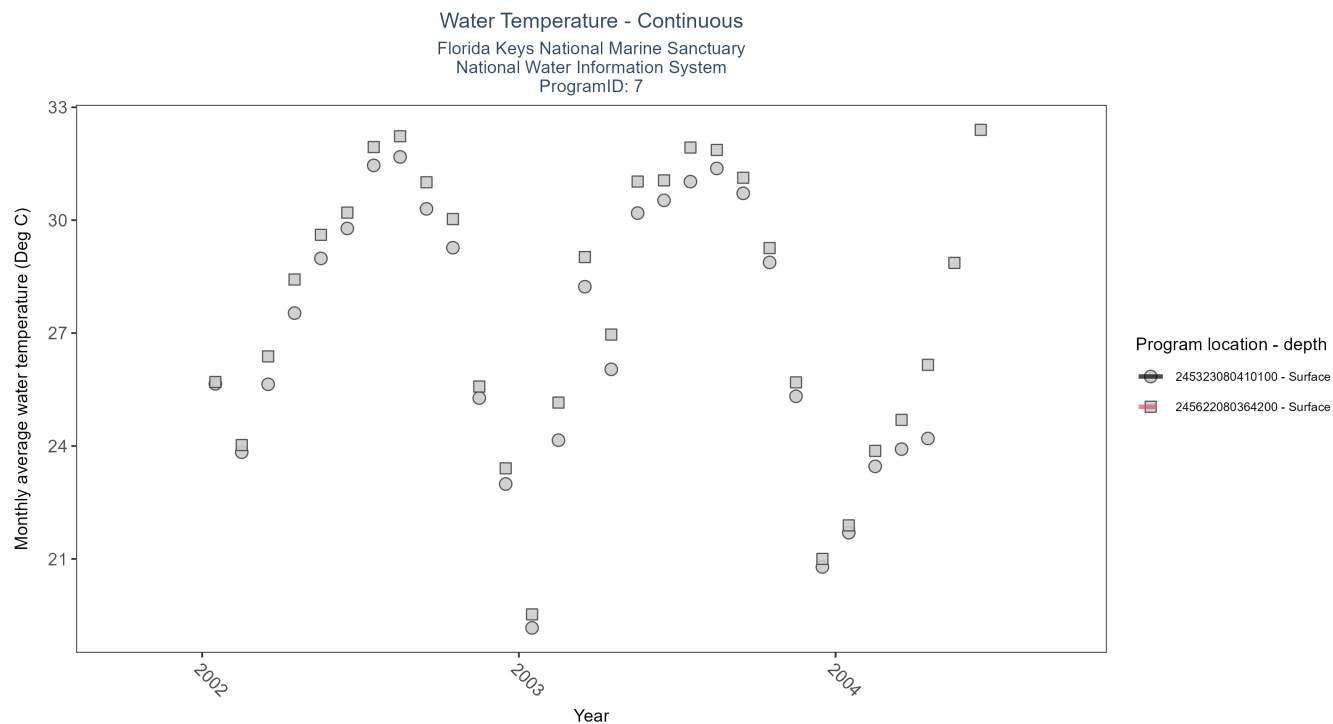


Figure 30: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Water Temperature - Program 7

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
245622080364200	Insufficient data to calculate trend	853	3	2002 - 2004	28.3	-	-	-	-
245323080410100	Insufficient data to calculate trend	791	3	2002 - 2004	27.9	-	-	-	-

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

Water Temperature - Continuous - Program 296

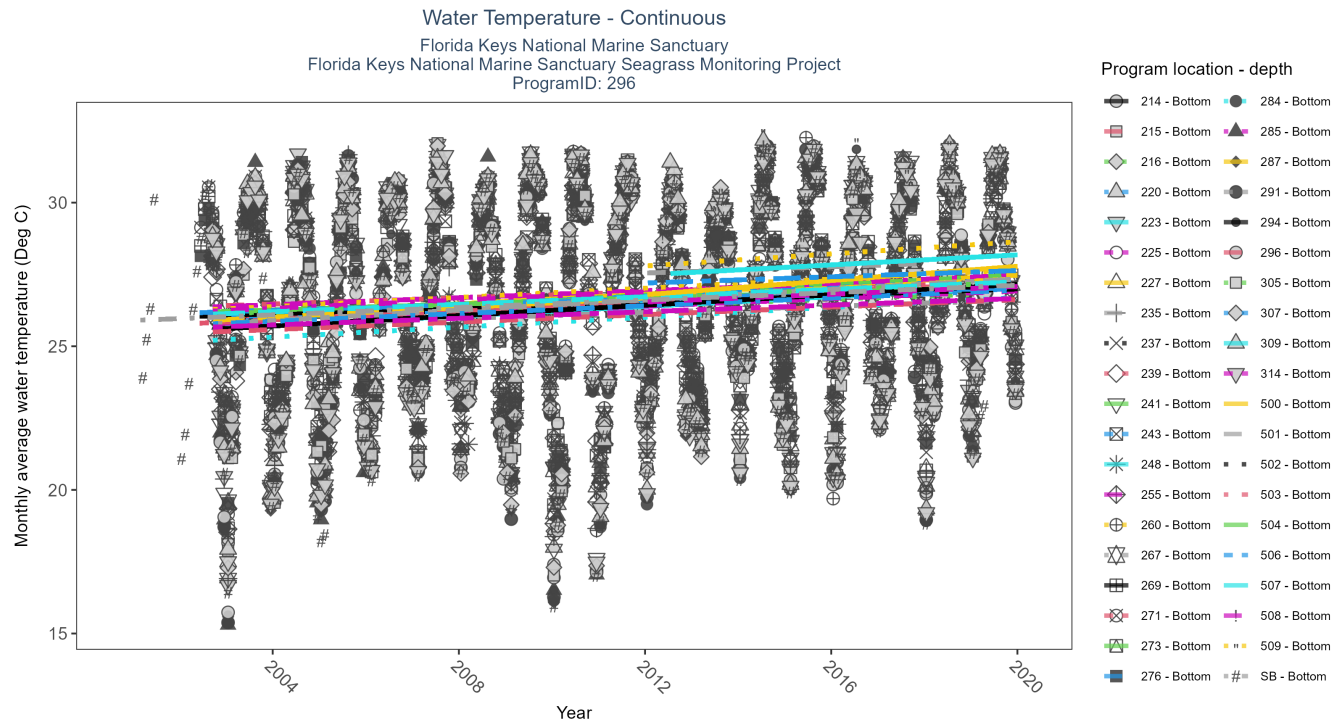


Figure 31: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 36: Seasonal Kendall-Tau Results for Water Temperature - Program 296

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
214	Significantly increasing trend	136333	18	2002 - 2019	26.52	0.27	25.84	0.07	0
215	Significantly increasing trend	133286	16	2003 - 2018	26.74	0.26	26.42	0.05	0
216	Significantly increasing trend	98535	17	2002 - 2018	26.26	0.31	25.86	0.06	0
220	Significantly increasing trend	126033	17	2003 - 2019	26.52	0.25	25.94	0.06	0
223	Significantly increasing trend	133082	18	2002 - 2019	26.89	0.3	25.84	0.08	0
225	Significantly increasing trend	117692	17	2002 - 2019	26.82	0.32	26.32	0.06	0
227	Significantly increasing trend	105351	17	2003 - 2019	26.67	0.29	26.06	0.08	0
235	Significantly increasing trend	128499	18	2002 - 2019	27.14	0.28	25.77	0.08	0
237	Significantly increasing trend	122250	18	2002 - 2019	26.38	0.31	25.74	0.09	0
239	Significantly increasing trend	111523	17	2002 - 2018	26.92	0.24	25.96	0.07	0
241	Significantly increasing trend	127914	18	2002 - 2019	27.26	0.27	25.91	0.09	0
243	Significantly increasing trend	121593	18	2002 - 2019	26.62	0.3	26	0.07	0
248	Significantly increasing trend	111702	18	2002 - 2019	26.79	0.31	25.54	0.08	0
255	Significantly increasing trend	119939	18	2002 - 2019	26.35	0.24	25.73	0.07	0
260	Significantly increasing trend	97832	16	2002 - 2019	27.07	0.28	26.22	0.08	0
267	Significantly increasing trend	99735	18	2002 - 2019	26.57	0.24	25.64	0.05	0
269	Significantly increasing trend	106458	17	2002 - 2019	26.74	0.21	26.02	0.05	0
271	Significantly increasing trend	133627	18	2002 - 2019	26.92	0.26	25.77	0.07	0
273	Significantly increasing trend	129817	18	2002 - 2019	27.16	0.24	26.16	0.05	0
276	Significantly increasing trend	123833	18	2002 - 2019	26.87	0.21	26.15	0.05	0
284	Significantly increasing trend	123977	17	2002 - 2019	26.86	0.28	25.14	0.09	0
285	Significantly increasing trend	121423	18	2002 - 2019	26.86	0.25	26.17	0.07	0
287	Significantly increasing trend	133008	18	2002 - 2019	26.87	0.29	25.84	0.08	0
291	Significantly increasing trend	116240	18	2002 - 2019	26.38	0.26	25.72	0.09	0
294	Significantly increasing trend	112348	18	2002 - 2019	26.92	0.27	25.52	0.09	0
296	Significantly increasing trend	114497	17	2002 - 2019	27.36	0.21	25.45	0.07	0
305	Significantly increasing trend	122296	18	2002 - 2019	26.43	0.22	26.07	0.06	0
307	Significantly increasing trend	110802	17	2002 - 2019	26.74	0.22	25.73	0.07	0
309	Significantly increasing trend	107410	18	2002 - 2019	27.85	0.27	26.07	0.06	0
314	Significantly increasing trend	110686	18	2002 - 2019	27.41	0.23	25.63	0.06	0
500	Significantly increasing trend	69048	8	2012 - 2019	27.33	0.23	26.79	0.12	0.01
501	No significant trend	34805	5	2012 - 2018	27.48	0.11	27.55	0.05	0.65
502	Insufficient data to calculate trend	22765	4	2016 - 2019	26.70	-	-	-	-
503	Insufficient data to calculate trend	7490	1	2016 - 2016	28.74	-	-	-	-
504	Insufficient data to calculate trend	4339	1	2018 - 2018	29.84	-	-	-	-
506	No significant trend	35198	7	2012 - 2019	27.41	0.04	27.2	0.05	0.74
507	No significant trend	47517	8	2012 - 2019	27.36	0.18	27.48	0.09	0.12
508	No significant trend	24021	6	2012 - 2019	26.67	0.33	26.54	0.07	0.29
509	No significant trend	38607	8	2012 - 2019	27.70	0.05	27.79	0.11	0.47
SB	Significantly increasing trend	145514	19	2001 - 2019	26.34	0.23	25.9	0.06	0

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

Water Temperature - Continuous - Program 899

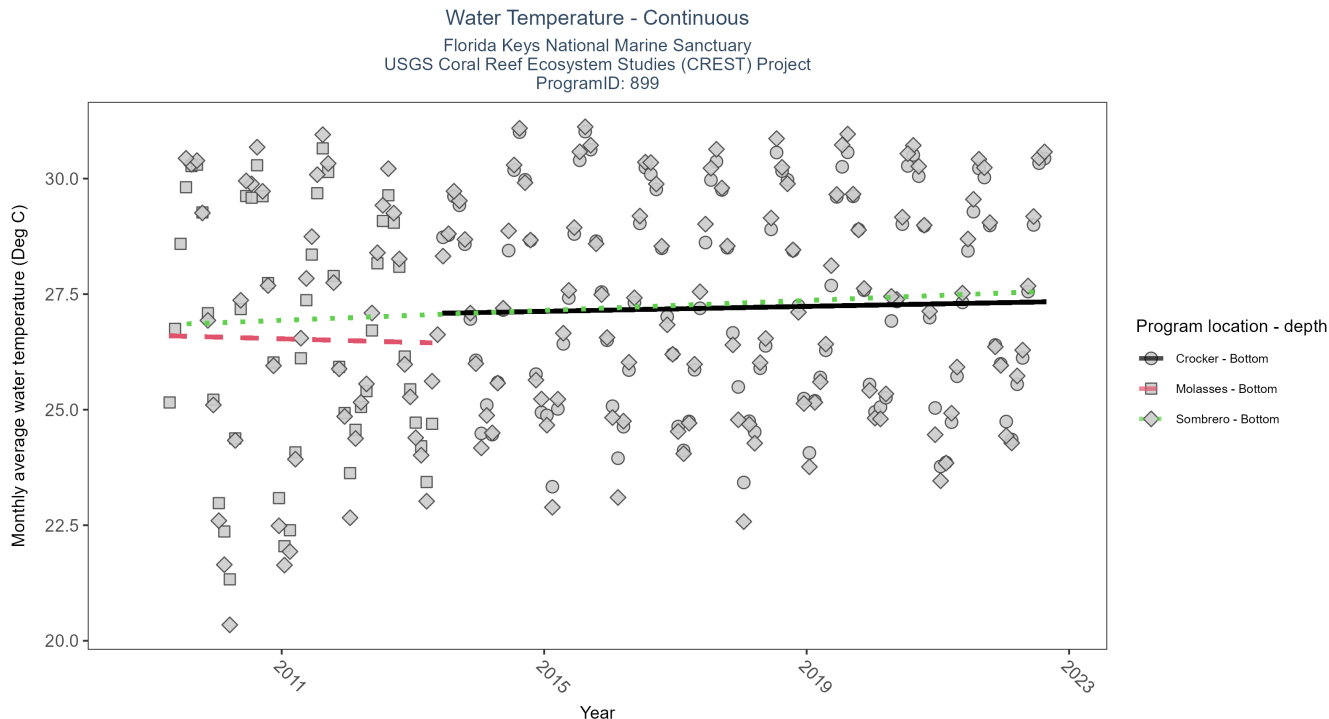


Figure 32: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 37: Seasonal Kendall-Tau Results for Water Temperature - Program 899

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Crocker	Significantly increasing trend	322670	10	2013 - 2022	27.32	0.15	27.07	0.03	0.04
Molasses	No significant trend	140713	5	2009 - 2013	26.72	-0.03	26.61	-0.04	0.92
Sombrero	Significantly increasing trend	459354	14	2009 - 2022	27.16	0.26	26.83	0.05	0.00

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

Water Temperature - Continuous - Program 986

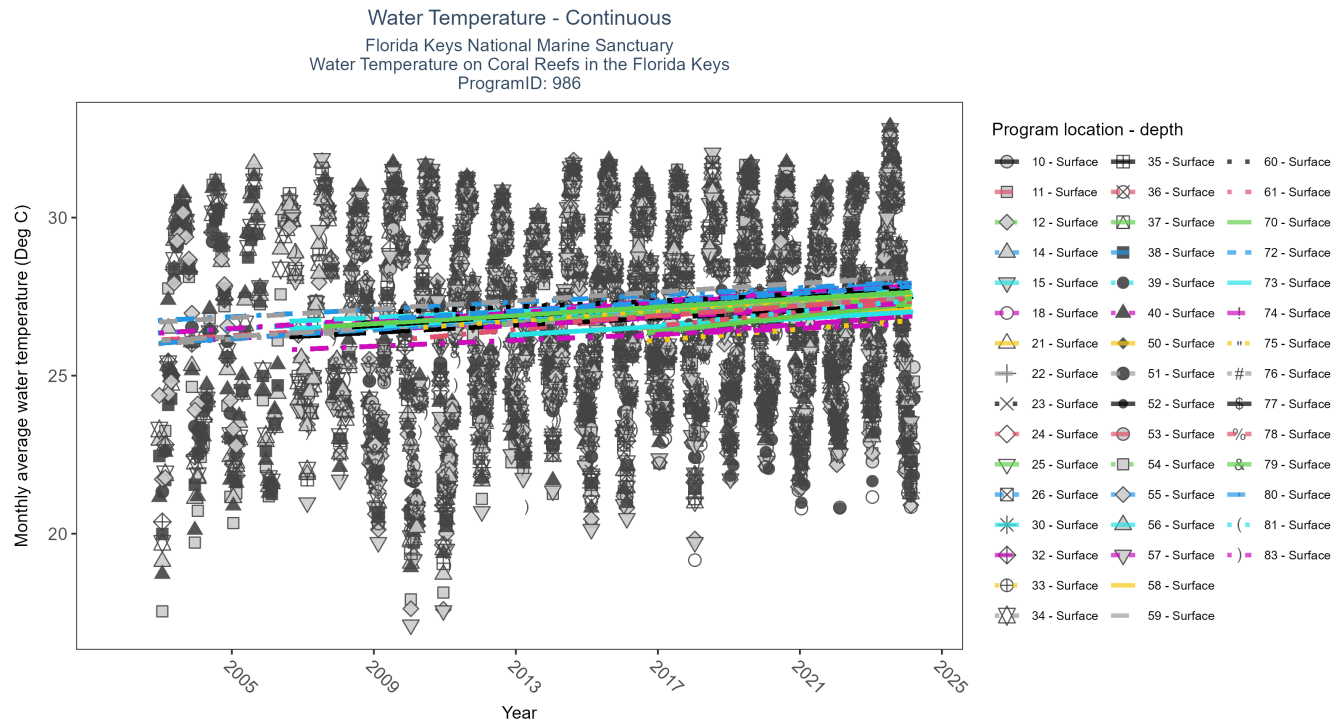


Figure 33: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 38: Seasonal Kendall-Tau Results for Water Temperature - Program 986

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
10	No significant trend	32760	5	2020 - 2024	27.48	0.23	26.68	0.20	0.18
11	Significantly increasing trend	243135	20	2003 - 2024	26.84	0.34	26.13	0.07	0.00
12	Significantly increasing trend	152036	15	2008 - 2024	27.18	0.26	26.41	0.07	0.00
14	Significantly increasing trend	238807	21	2002 - 2024	26.96	0.31	26.29	0.06	0.00
15	Significantly increasing trend	218855	19	2006 - 2024	27.04	0.23	26.45	0.06	0.00
18	Significantly increasing trend	59983	9	2016 - 2024	27.18	0.20	26.21	0.08	0.03
21	Significantly increasing trend	69835	9	2016 - 2024	27.23	0.20	26.46	0.08	0.03
22	Significantly increasing trend	185473	16	2009 - 2024	26.96	0.28	26.35	0.07	0.00
23	Significantly increasing trend	128184	13	2012 - 2024	27.43	0.27	26.73	0.08	0.00
24	Significantly increasing trend	126389	13	2010 - 2024	27.08	0.40	26.16	0.10	0.00
25	Significantly increasing trend	132181	14	2010 - 2024	27.28	0.21	26.73	0.07	0.00
26	Significantly increasing trend	156891	16	2009 - 2024	27.08	0.31	26.74	0.08	0.00
30	Significantly increasing trend	131074	13	2012 - 2024	26.67	0.26	26.23	0.06	0.00
32	Significantly increasing trend	237317	20	2003 - 2024	26.74	0.36	26.04	0.07	0.00
33	No significant trend	50595	8	2016 - 2024	27.06	0.17	26.04	0.09	0.18
34	Significantly increasing trend	289051	23	2002 - 2024	26.84	0.36	25.94	0.07	0.00
35	Significantly increasing trend	232782	19	2006 - 2024	26.98	0.28	26.19	0.06	0.00
36	Significantly increasing trend	207795	18	2007 - 2024	27.01	0.32	26.43	0.07	0.00
37	Significantly increasing trend	67333	9	2016 - 2024	27.03	0.30	26.27	0.12	0.00
38	Significantly increasing trend	271179	23	2002 - 2024	26.59	0.35	25.96	0.07	0.00
39	No significant trend	48939	7	2018 - 2024	27.06	0.09	27.06	0.08	0.45
40	Significantly increasing trend	259048	23	2002 - 2024	26.89	0.33	26.28	0.07	0.00
50	Significantly increasing trend	118231	12	2013 - 2024	27.11	0.33	26.82	0.07	0.00
51	Significantly increasing trend	237296	20	2003 - 2024	26.74	0.37	26.11	0.06	0.00
52	Significantly increasing trend	202154	17	2008 - 2024	27.01	0.40	26.61	0.08	0.00
53	Significantly increasing trend	193391	17	2008 - 2024	27.06	0.42	26.53	0.08	0.00
54	Significantly increasing trend	142926	13	2012 - 2024	27.09	0.31	26.64	0.06	0.00
55	Significantly increasing trend	231874	23	2002 - 2024	26.94	0.33	26.69	0.06	0.00
56	Significantly increasing trend	190431	19	2006 - 2024	26.81	0.28	26.69	0.05	0.00
57	Significantly increasing trend	202888	17	2008 - 2024	27.08	0.38	26.61	0.08	0.00
58	Significantly increasing trend	87137	11	2014 - 2024	27.35	0.25	26.94	0.07	0.00
59	Significantly increasing trend	206603	20	2002 - 2024	26.96	0.37	26.63	0.07	0.00
60	Significantly increasing trend	156278	16	2009 - 2024	27.01	0.23	26.99	0.05	0.00
61	Significantly increasing trend	68996	9	2016 - 2024	27.35	0.39	26.46	0.11	0.00
70	Significantly increasing trend	119048	12	2013 - 2024	27.01	0.33	26.73	0.06	0.00
72	Significantly increasing trend	198393	16	2008 - 2023	26.86	0.47	26.39	0.08	0.00
73	Significantly increasing trend	193376	17	2008 - 2024	26.81	0.40	26.47	0.08	0.00
74	Significantly increasing trend	142860	13	2012 - 2024	26.94	0.31	26.52	0.06	0.00
75	Significantly increasing trend	159540	15	2010 - 2024	27.23	0.39	26.49	0.08	0.00
76	Significantly increasing trend	183695	16	2009 - 2024	26.98	0.35	26.63	0.07	0.00
77	Significantly increasing trend	203309	17	2008 - 2024	27.01	0.36	26.51	0.07	0.00
78	Significantly increasing trend	102736	11	2014 - 2024	27.16	0.27	26.66	0.08	0.00
79	Significantly increasing trend	190275	18	2007 - 2024	26.94	0.33	26.51	0.07	0.00
80	Significantly increasing trend	182341	16	2009 - 2024	27.03	0.33	26.77	0.07	0.00
81	Significantly increasing trend	68931	9	2016 - 2024	27.35	0.39	26.48	0.11	0.00
83	Significantly increasing trend	139311	17	2006 - 2023	25.86	0.19	25.79	0.05	0.00

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

Water Temperature - Continuous - Program 989

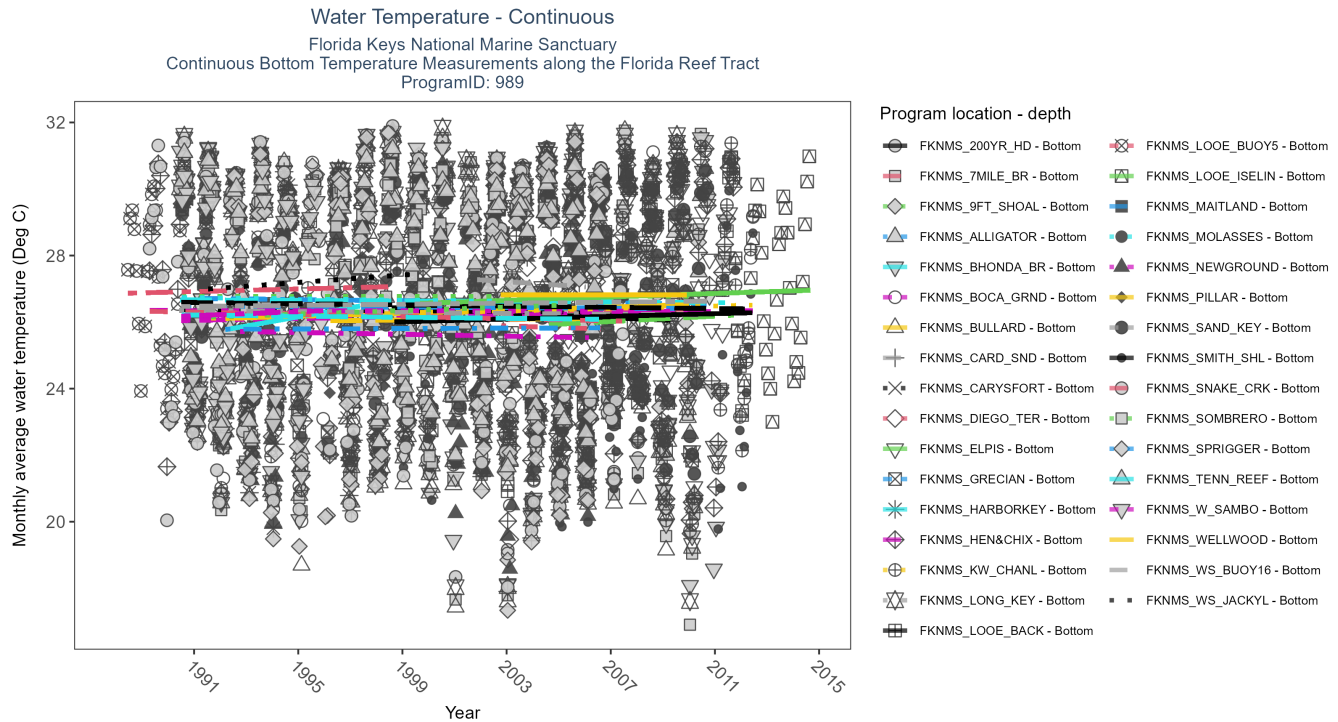


Figure 34: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 39: Seasonal Kendall-Tau Results for Water Temperature - Program 989

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKNMS-200YR-HD	No significant trend	44601	12	1998 - 2009	26.10	-0.1	26.45	-0.04	0.17
FKNMS-7MILE-BR	No significant trend	73055	19	1991 - 2010	26.66	0.05	26.22	0.01	0.35
FKNMS-9FT-SHOAL	No significant trend	80299	21	1990 - 2010	26.50	0	26.76	0	0.99
FKNMS-ALLIGATOR	No significant trend	65144	19	1990 - 2010	26.55	-0.06	26.72	-0.01	0.23
FKNMS-BHONDA-BR	No significant trend	77111	22	1990 - 2011	26.60	-0.02	26.67	0	0.66
FKNMS-BOCA-GRND	No significant trend	73434	17	1990 - 2012	26.14	0.08	26.04	0.01	0.17
FKNMS-BULLARD	Significantly increasing trend	66230	18	1992 - 2009	26.31	0.12	26.11	0.02	0.03
FKNMS-CARD-SND	No significant trend	18249	6	2001 - 2006	26.52	-0.05	27.32	-0.05	0.79
FKNMS-CARYSFORT	No significant trend	55001	16	1990 - 2006	26.40	-0.03	26.38	0	0.64
FKNMS-DIEGO-TER	No significant trend	16693	5	2002 - 2006	25.58	-0.05	25.91	-0.03	0.84
FKNMS-ELPIS	No significant trend	31035	8	2004 - 2011	26.35	0.06	25.9	0.04	0.53
FKNMS-GRECIAN	No significant trend	51723	18	1990 - 2010	26.65	-0.03	26.48	0	0.66
FKNMS-HARBORKEY	No significant trend	15407	5	1992 - 1997	26.50	0.14	25.74	0.14	0.33
FKNMS-HEN-and-CHIX	No significant trend	72285	21	1989 - 2011	26.50	-0.01	26.35	0	0.88
FKNMS-KW-CHANL	No significant trend	123578	18	1991 - 2012	26.27	0.1	26.11	0.02	0.08
FKNMS-LONG-KEY	No significant trend	69656	19	1990 - 2010	26.64	-0.03	26.35	-0.01	0.58
FKNMS-LOOE-BACK	No significant trend	84984	18	1990 - 2012	26.80	-0.06	26.6	-0.01	0.42
FKNMS-LOOE-BUOY5	No significant trend	35252	10	1988 - 1998	26.90	0.05	26.86	0.02	0.36
FKNMS-LOOE-ISELIN	No significant trend	194367	13	1999 - 2014	26.88	0.13	26.55	0.03	0.08
FKNMS-MAITLAND	Insufficient data to calculate trend	12421	4	2004 - 2007	26.07	-	-	-	-
FKNMS-MOLASSES	No significant trend	36146	13	1990 - 2002	26.70	-0.05	26.74	-0.01	0.48
FKNMS-NEWGROUND	No significant trend	35329	12	1992 - 2006	25.49	-0.05	25.73	-0.01	0.52
FKNMS-PILLAR	No significant trend	40805	11	1996 - 2006	26.24	0.02	26.04	0.01	0.94
FKNMS-SAND-KEY	No significant trend	59287	18	1990 - 2010	26.70	0.05	26.46	0.01	0.32
FKNMS-SMITH-SHL	No significant trend	94527	10	1998 - 2012	25.45	0.13	25.99	0.02	0.19
FKNMS-SNAKE-CRK	No significant trend	56777	19	1989 - 2007	26.16	-0.06	26.33	-0.02	0.28
FKNMS-SOMBRERO	No significant trend	48974	13	1991 - 2005	26.50	0.13	26.14	0.03	0.05
FKNMS-SPRIGGER	No significant trend	41834	13	1992 - 2006	26.10	0.02	25.78	0	0.86
FKNMS-TENN-REEF	No significant trend	63260	16	1990 - 2006	26.70	-0.06	26.22	-0.01	0.27
FKNMS-WELLWOOD	No significant trend	30427	8	2002 - 2009	26.43	0	26.82	0	1
FKNMS-WS-BUOY16	Insufficient data to calculate trend	8123	3	2003 - 2005	25.99	-	-	-	-
FKNMS-WS-JACKYL	No significant trend	29557	9	1991 - 1999	26.40	0.17	26.96	0.06	0.09
FKNMS-W-SAMBO	No significant trend	18786	6	1990 - 1995	26.90	0.09	26.16	0.03	0.56

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient

data to fit a model for nine locations.

Water Temperature - Continuous - Program 10004

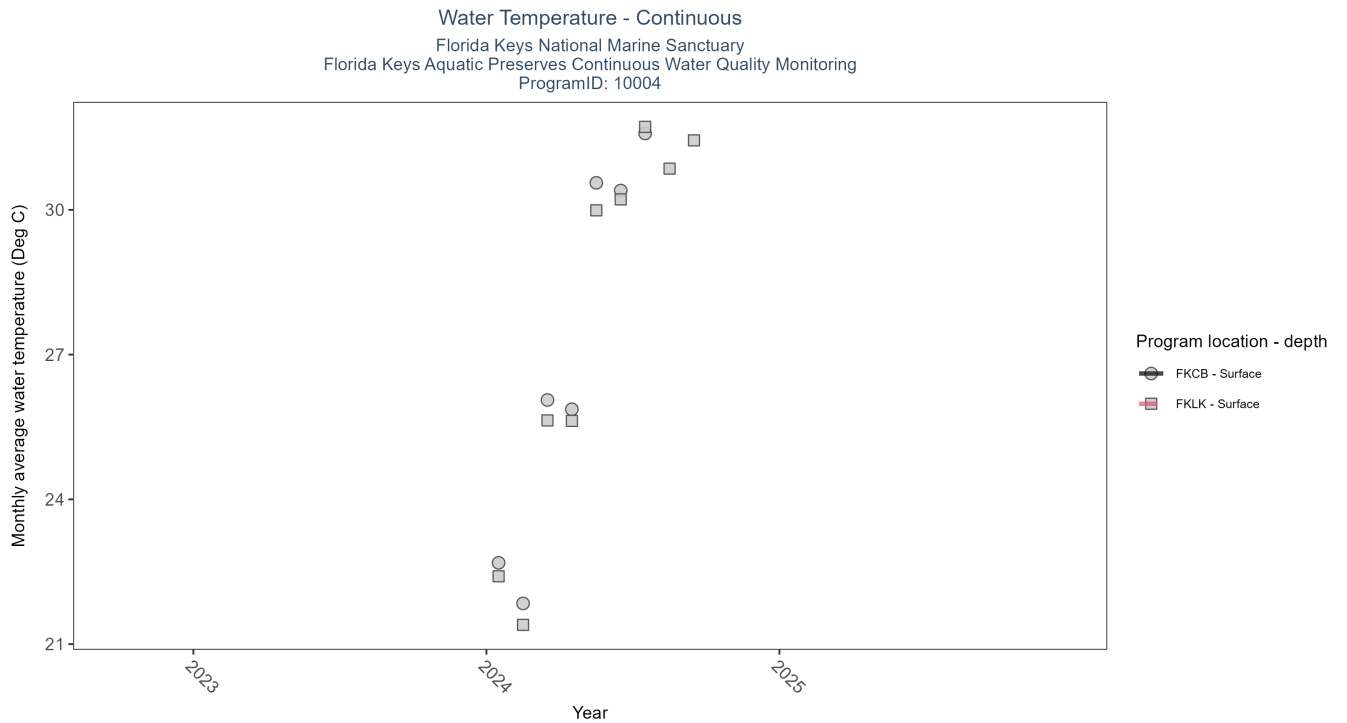


Figure 35: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 40: Seasonal Kendall-Tau Results for Water Temperature - Program 10004

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16263	1	2024 - 2024	26.8	-	-	-	-
FKLK	Insufficient data to calculate trend	21517	1	2024 - 2024	29.0	-	-	-	-

At eighty-two program locations, monthly average water temperature increased between 0.01 and 0.16°C per year. No detectable change in monthly average water temperature was observed at forty-one locations. There was insufficient data to fit a model for nine locations.

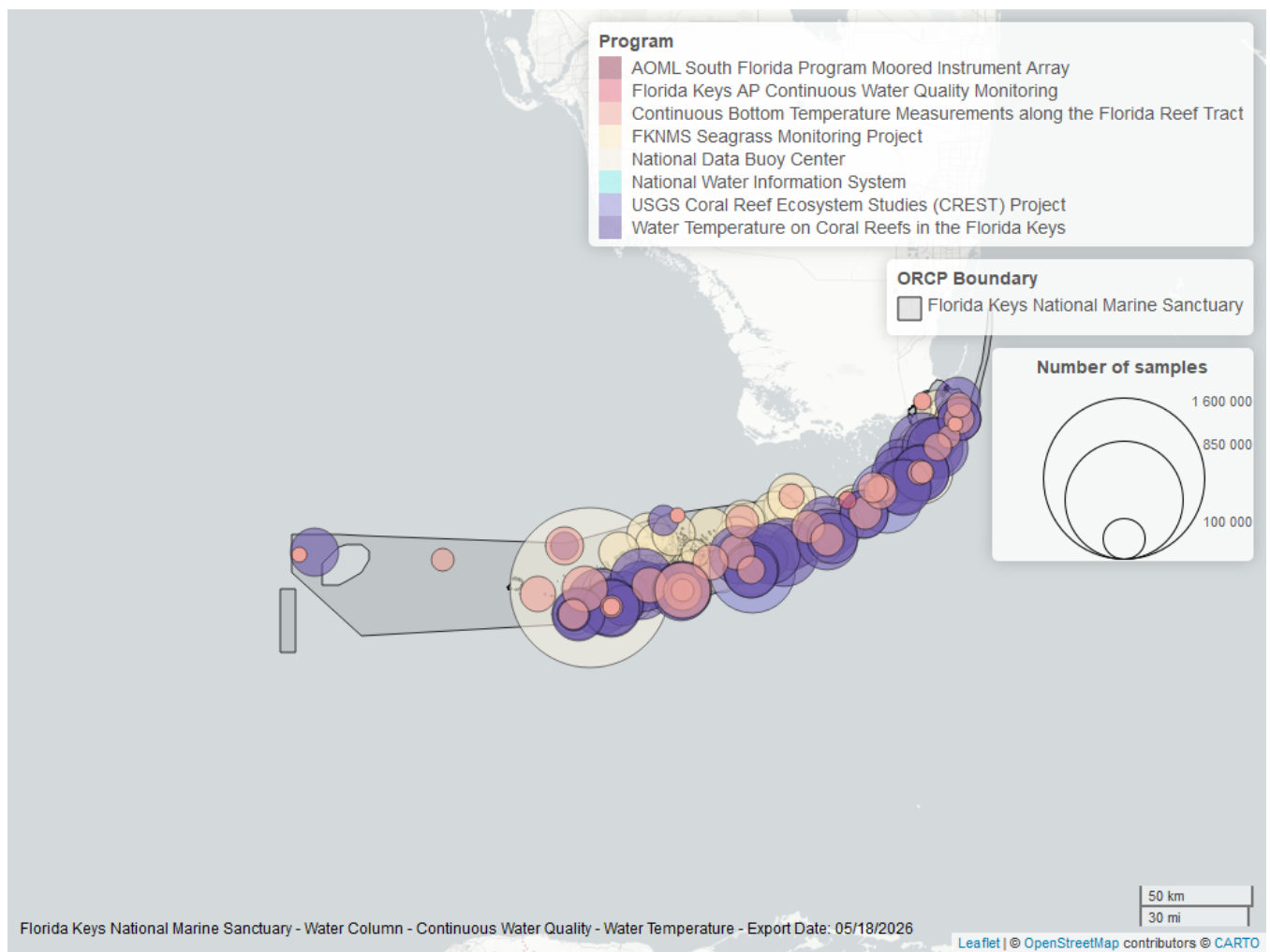


Figure 36: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous



Figure 37: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 41: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16258	1	2024 - 2024	36.50	-	-	-	-
1B	No significant trend	86204	6	2003 - 2008	36.07	0.24	35.68	0.17	0.05
245622080364200	Insufficient data to calculate trend	764	3	2002 - 2004	35.00	-	-	-	-
245323080410100	Insufficient data to calculate trend	746	3	2002 - 2004	35.00	-	-	-	-
FKLK	Insufficient data to calculate trend	21517	1	2024 - 2024	36.10	-	-	-	-

No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for four locations.

Specific Conductivity - Continuous

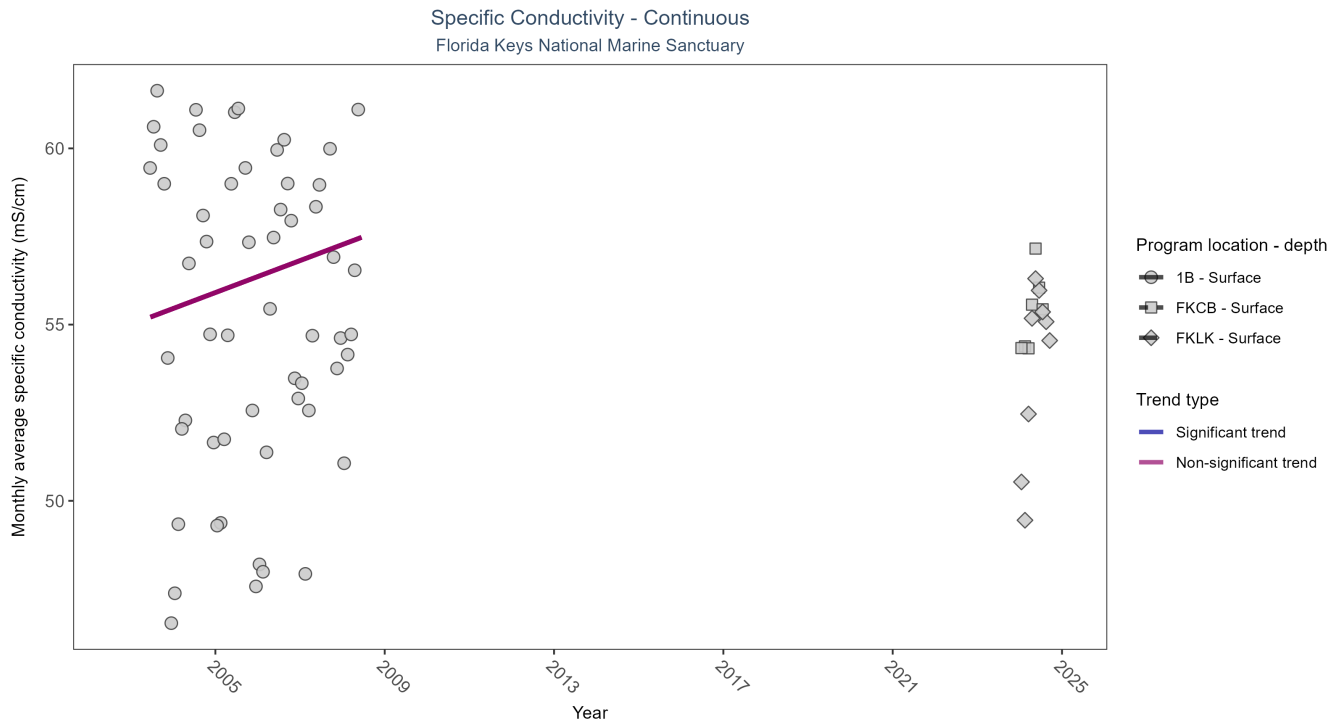


Figure 38: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 42: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16258	1	2024 - 2024	55.12	-	-	-	-
1B	No significant trend	86204	6	2003 - 2008	56.33	0.21	55	0.45	0.1
FKLK	Insufficient data to calculate trend	21517	1	2024 - 2024	54.71	-	-	-	-

No detectable change in monthly average specific conductivity was observed at one location. There was insufficient data to fit a model for two locations.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida’s estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson’s seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson’s seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Florida Keys National Marine Sanctuary
SAV Percent Cover - Sample Locations

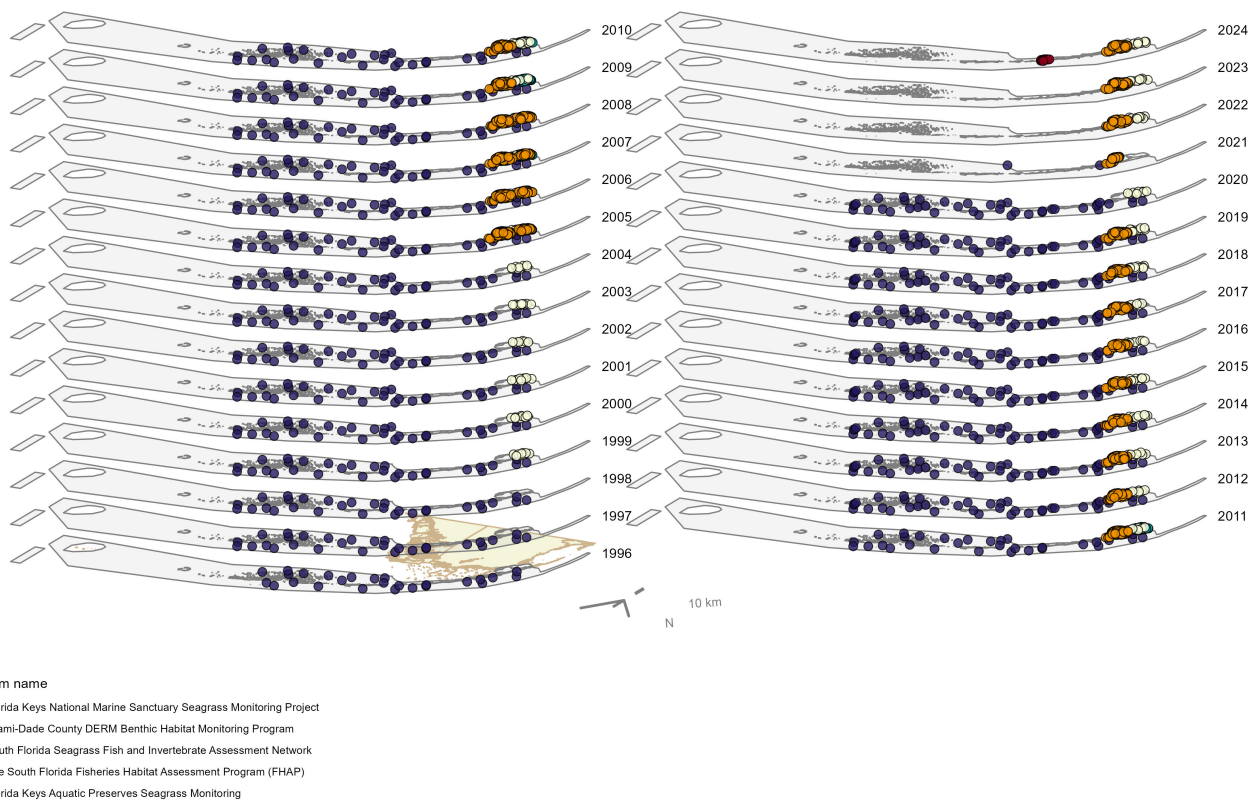


Figure 39: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Florida Keys National Marine Sanctuary* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

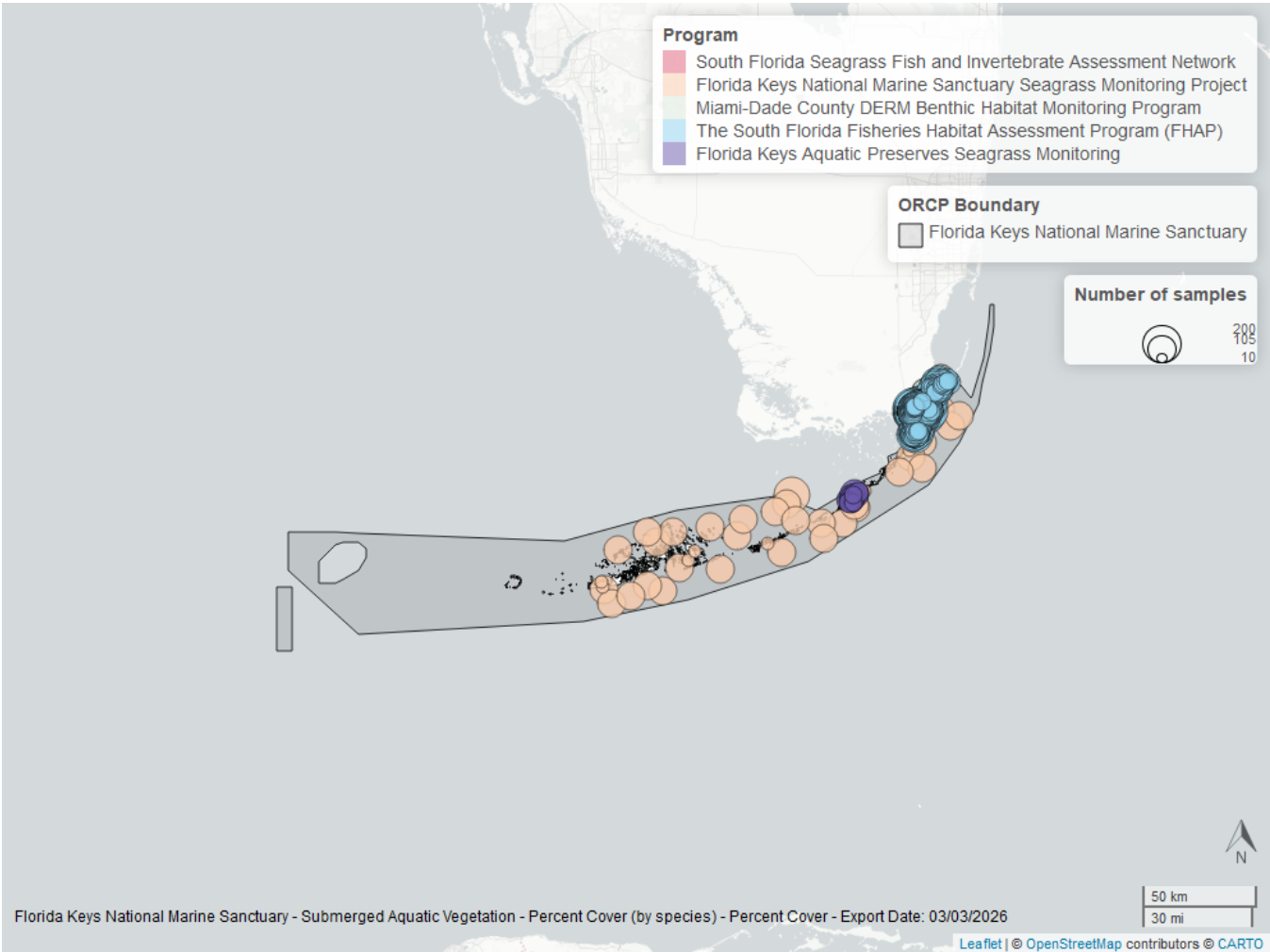


Figure 40: Map showing SAV sampling sites within the boundaries of *Florida Keys National Marine Sanctuary*. The point size reflects the number of samples at a given sampling site.

Table 43: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
296	4200	1996	2021	Braun Blanquet	40
965	65538	2005	2011	Braun Blanquet	87
4018	4328	1999	2024	Braun Blanquet	120
4049	109886	2005	2025	Braun Blanquet	1333
10007	382	2024	2025	Braun Blanquet	16
4018	279	1999	2007	Percent Cover	67
10007	785	2024	2025	Percent Cover	16

Program names:

296 - Florida Keys National Marine Sanctuary Seagrass Monitoring Project²³

965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹⁸

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³
 4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹²
 10007 - Florida Keys Aquatic Preserves Seagrass Monitoring[[@](#)]

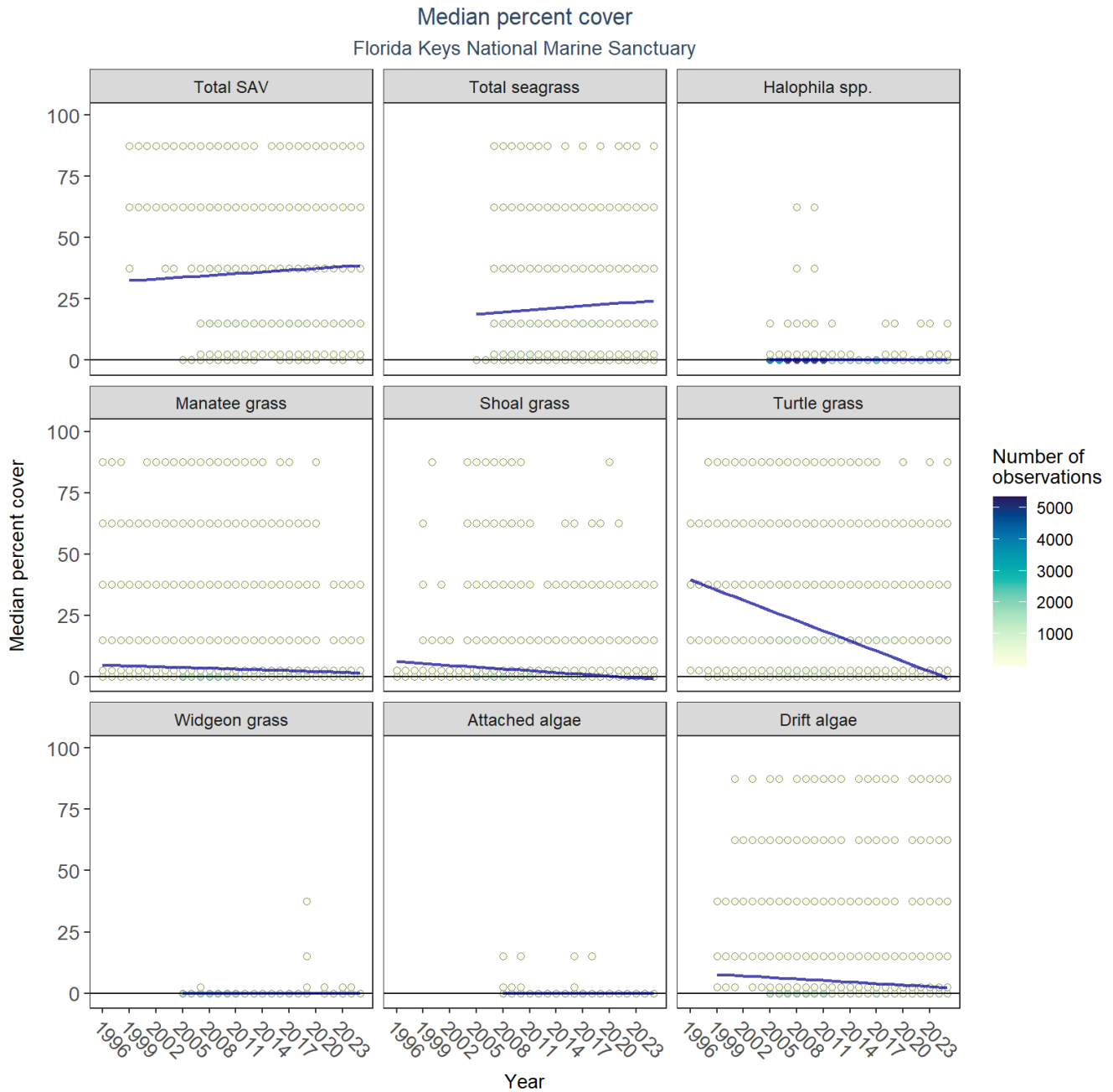


Figure 41: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

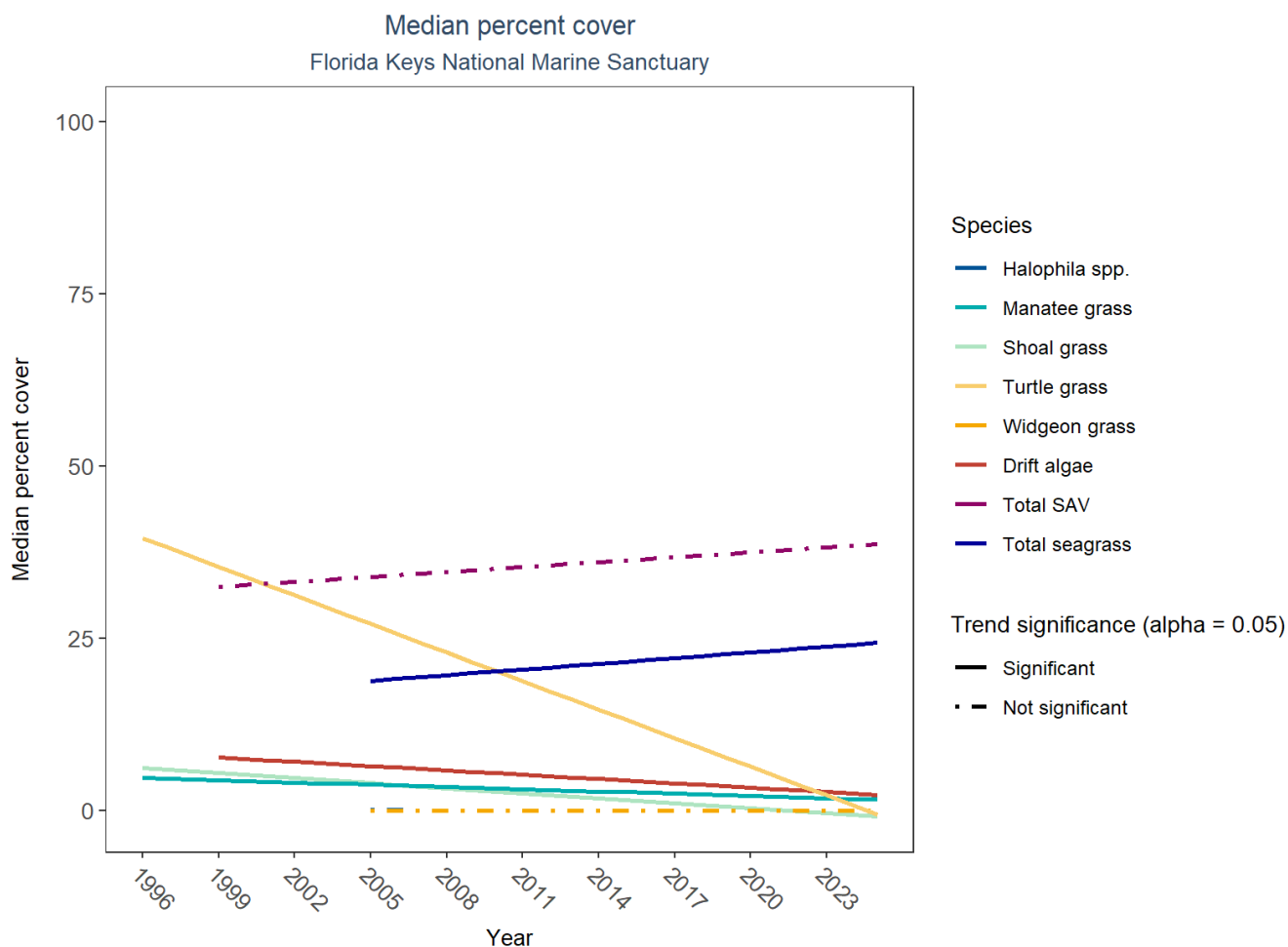


Figure 42: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 44: Percent Cover Trend Analysis for Florida Keys National Marine Sanctuary

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Attached algae	No detectable trend	2008 - 2025	0.0741863	-0.0024491	0.1967401
Drift algae	Decreasing trend	1999 - 2025	8.7431957	-0.2077773	0.0000786
Shoal grass	Decreasing trend	1996 - 2025	6.6589301	-0.2440056	0.0002912
Halophila spp.	No detectable trend	2005 - 2025	0.0693825	-0.0018662	0.4135004
Widgeon grass	No detectable trend	2005 - 2025	-0.0286063	0.0022047	0.1272600
Manatee grass	Decreasing trend	1996 - 2025	4.9412447	-0.1079911	0.0175022
Turtle grass	Decreasing trend	1996 - 2025	42.2873518	-1.3820602	0.0000000
Total SAV	No detectable trend	1999 - 2025	31.2798285	0.2376571	0.1166578
Total seagrass	Increasing trend	2005 - 2025	15.7746961	0.2765583	0.0010334

An annual increase in percent cover was observed for total seagrass (0.28%). Annual decreases in percent cover were observed for manatee grass (-0.11%), shoal grass (-0.24%), turtle grass (-1.38%), and drift algae (-0.21%). Total SAV, Halophila spp., widgeon grass, and attached algae showed no detectable change in percent cover.

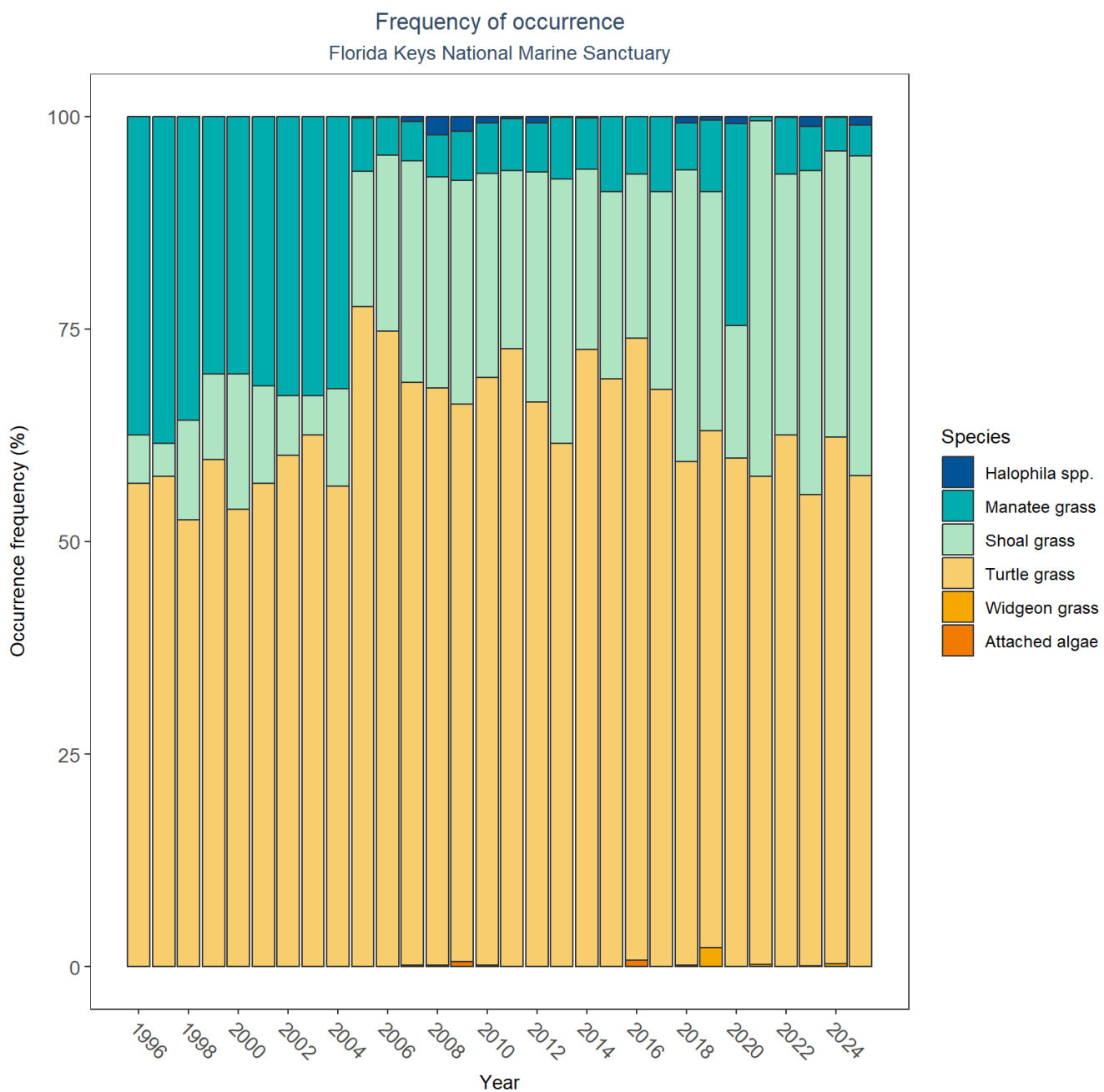


Figure 43: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Florida Keys National Marine Sanctuary within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Coral Reef

The data file used is: **All_CORAL_Parameters-2026-Jun-04.txt**

Percent Cover

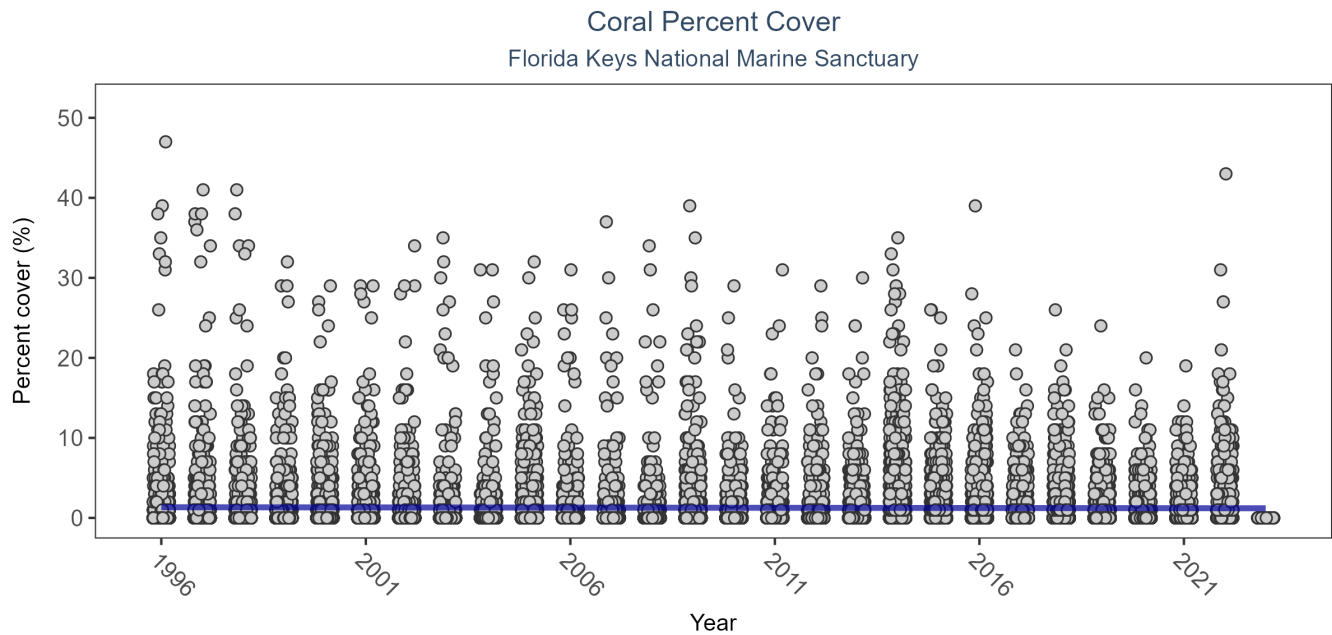


Figure 44: Scatter plot of live coral coverage over time as a percent of reef surface. Species groups include octocorals, milleporans, and scleractinians. If the time series included five or more years of observations, a significant (blue) or non-significant (magenta) trend line is also shown. Data points are jittered horizontally to reduce overlap.

Table 45: Coral Percent Cover

Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Decreasing trend	1996 - 2023	11.0811	-0.00488	0

Annual average percent cover decreased by less than 0.01% between 1996 and 2023.

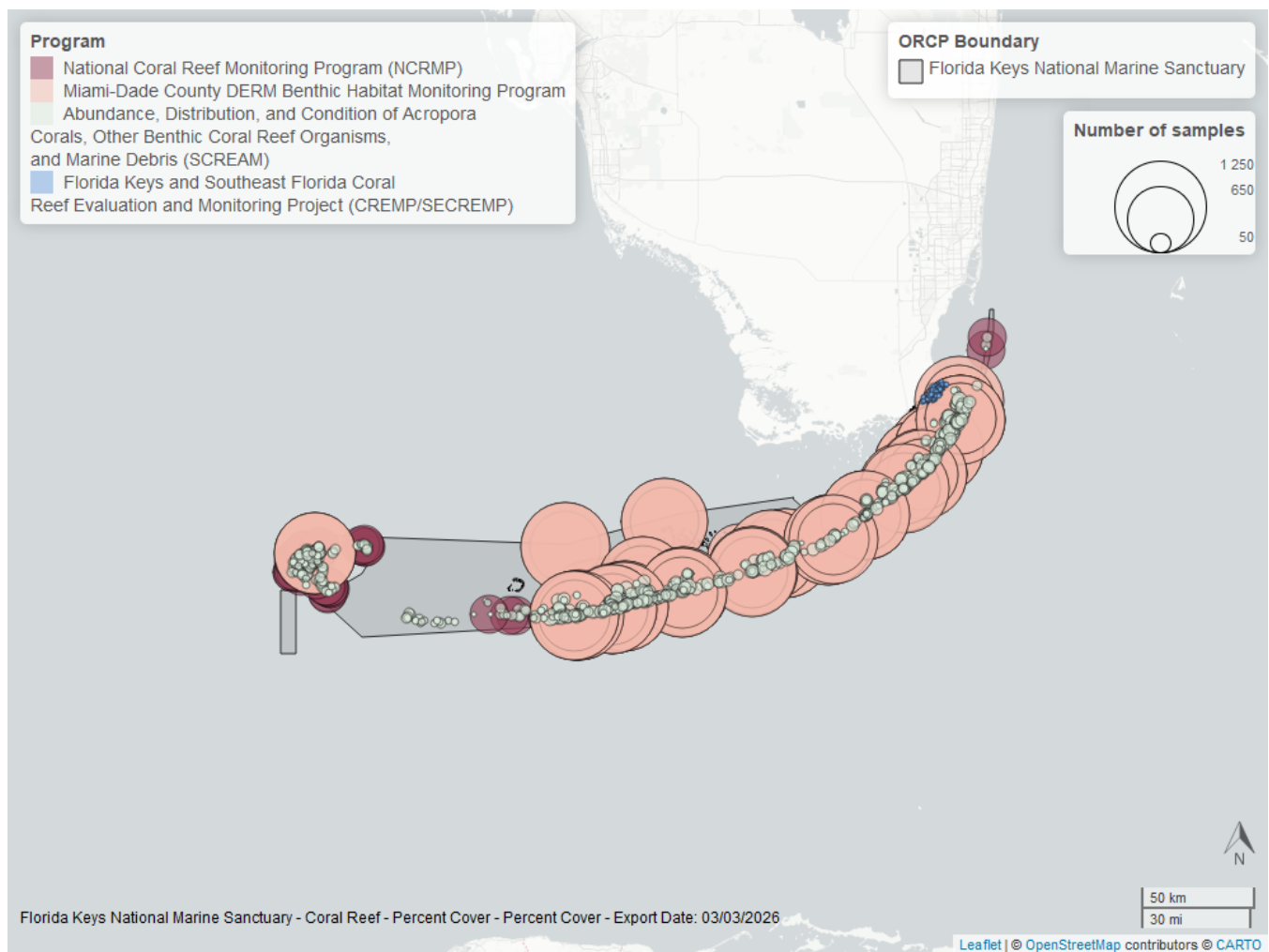


Figure 45: Map showing location of coral percent cover sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species Richness

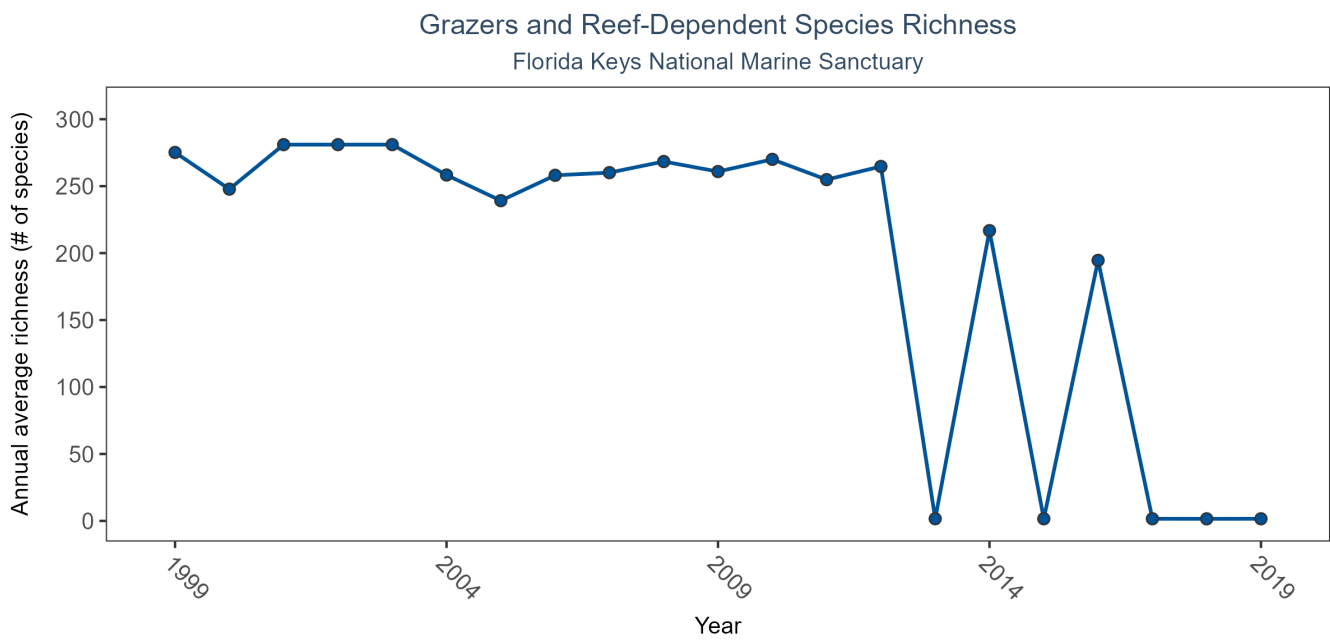


Figure 46: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.

Table 46: Coral Species Richness

<i>No. of Samples</i>	<i>No. Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
9898	21	1999 - 2019	281	248.1485

The median annual number of taxa was 281 based on 9,898 observations collected between 1999 and 2019.

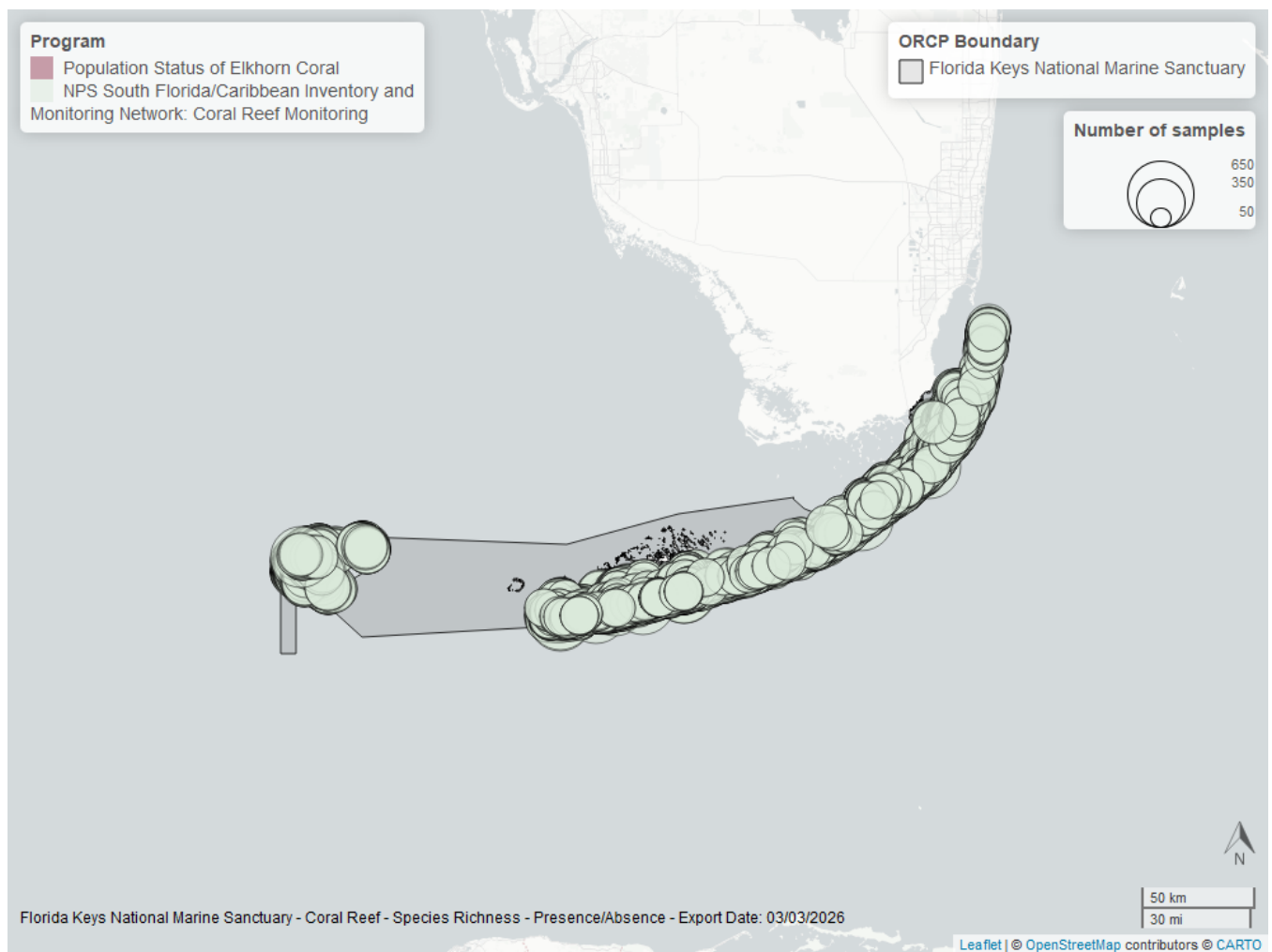


Figure 47: Map showing location of coral species richness sampling locations within the boundaries of *Florida Keys National Marine Sanctuary*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Abudefdud saxatilis ²	Epinephelus itajara ²	Padina gymnospora ¹
Acanthemblemaria aspera ²	Epinephelus morio ²	Pagrus pagrus ²
Acanthemblemaria chaplini ²	Epinephelus striatus ²	Palythoa mammosa
Acanthemblemaria maria ²	Eques lanceolatus ²	Palythoa spp.
Acanthemblemaria spinosa ²	Equetus punctatus ²	Pandaros acanthifolium
Acanthocybium solandri	Ernodesmis sp. ¹	Parablennius marmoreus ²
Acanthophora muscoides ¹	Erylus formosus	Paraclinus marmoratus ²
Acanthophora sp. ¹	Erythropodium caribaeorum ³	Paraclinus nigripinnis ²
Acanthostracion polygonium ²	Eucinostomus argenteus	Paralichthys albigutta ²
Acanthostracion quadricornis ²	Eucinostomus gula	Paranthias furcifer ²
Acanthurus bahianus ²	Eucinostomus jonesii	Pareques acuminatus ²
Acanthurus chirurgus ²	Eunicea calyculata ³	Pareques umbrosus ²
Acanthurus coeruleus ²	Eunicea flexuosa ³	Pempheris schomburgkii ²
Acanthurus sp. ²	Eunicea fusca ³	Penaeus monodon
Acetabularia sp. ¹	Eunicea knighti ³	Penicillus sp. ¹
Acetabularia spp. ¹	Eunicea laciniata ³	Penicillus spp. ¹
Acropora cervicornis ³	Eunicea laxispica ³	Peyssonnelia
Acropora palmata ³	Eunicea mammosa ³	Phaeoptyx xenus ²
Acropora prolifera ³	Eunicea palmeri ³	Phorbas sp.
Actiniaria	Eunicea succinea ³	Phyllangia americana ³
Aetobatus narinari	Eunicea tourneforti ³	Phymanthus crucifer
Agardhiella ramosissima ¹	Eusmilia fastigiata ³	Plakortis angulospiculatus
Agaricia agaricites ³	Euthynnus alletteratus	Plexaura homomalla ³
Agaricia fragilis ³	Favia fragum ³	Plexaura kuna ³
Agaricia grahamae ³	Fine turf	Plexaurella dichotoma ³
Agaricia humilis ³	Fistularia tabacaria ²	Plexaurella grandiflora ³
Agaricia lamarcki ³	Fowlerichthys ocellatus ²	Plexaurella grisea ³
Agaricia sp.	Galaxaura spp.	Plexaurella nutans ³
Agaricia spp. ³	Galeocerdo cuvier ²	Polysiphonia sp. ¹
Agaricia tenuifolia	Geodia gibberosa	Pomacanthus arcuatus ²
Agaricia undata ³	Geodia neptuni	Pomacanthus paru ²
Agelas clathrodes	Gerres cinereus	Porifera
Agelas conifera	Ginglymostoma cirratum	Porifera spp.
Agelas dispar	Gnatholepis thompsoni ²	Porites astreoides ³
Agelas schmidtii	Gobioclinus bucciferus ²	Porites branneri ³
Agelas wiedenmayeri	Gobioclinus filamentosus ²	Porites cf. branneri
Ahlia egmontis	Gobioclinus gobio ²	Porites colonensis ³
Aiolochoira crassa	Gobioclinus kalisherae ²	Porites divaricata ³
Albula vulpes	Gobiosoma sp. ²	Porites furcata ³
Alcyonacea sp. ³	Gorgonia flabellum ³	Porites porites ³
Alectis ciliaris ²	Gorgonia mariae ³	Porites sp.
Alphestes afer ²	Gorgonia ventalina ³	Porites spp. ³
Aluterus monoceros ²	Gracilaria sp. ¹	Priacanthus arenatus ²
Aluterus schoepfii ²	Gramma loreto ²	Priolepis hipoliti ²
Aluterus scriptus ²	Grateloupia ¹	Prionotus ophryus ²
Aluterus sp. ²	Griffithsia ¹	Prionotus rubio ²
Amblycirrhitis pinos ²	Gymnothorax funebris ²	Pristipomoides aquilonaris ²
Amphimedon compressa	Gymnothorax miliaris ²	Pristis pectinata
Amphimedon viridis	Gymnothorax moringa ²	Prognathodes aculeatus ²
Amphiroa spp. ¹	Gymnothorax nigromarginatus ²	Pseudobatos lentiginosus
Anadyomene linkiana ¹	Gymnothorax saxicola ²	Pseudodiploria clivosa ³
Anadyomene spp. ¹	Gymnothorax vicinus ²	Pseudodiploria strigosa ³
Anadyomene stellata ¹	Haemulon album ²	Pseudoplexaura crucis ³
Anchoa lyolepis	Haemulon aurolineatum ²	Pseudoplexaura flagellosa ³

Anisotremus surinamensis²
 Anisotremus virginicus²
 Antillogorgia acerosa³
 Antillogorgia americana³
 Antillogorgia bipinnata³
 Antillogorgia kallos³
 Antillogorgia rigida³
 Aplysina archeri
 Aplysina cauliformis
 Aplysina fistularis
 Aplysina fulva
 Aplysina lacunosa
 Apogon aurolineatus²
 Apogon binotatus²
 Apogon maculatus²
 Apogon phenax²
 Apogon pseudomaculatus²
 Apogon quadrisquamatus²
 Apogon townsendi²
 Archosargus probatocephalus²
 Archosargus rhomboidalis²
 Arturia canariensis
 Ascidiacea
 Astrapogon puncticulatus²
 Astrapogon sp.²
 Astrapogon stellatus²
 Astroscopus guttatus
 Atherinomorus stipes
 Aulostomus maculatus²
 Avrainvillea¹
 Avrainvillea levis¹
 Axinellida
 Azurina cyanea²
 Balistes capricus²
 Balistes sp.²
 Balistes vetula²
 Bare substrate
 Bartholomea annulata
 Batophora oerstedii¹
 Batophora spp.¹
 Blenniidae sp.²
 Bodianus pulchellus²
 Bodianus rufus²
 Bollmannia boqueronensis²
 Bothus lunatus²
 Bothus ocellatus²
 Brachygenys chrysargyreum²
 Branching gorgonian³
 Briareum asbestinum³
 Brockius nigricinctus²
 Brotula barbata
 Brown algae¹
 Bryopsis¹
 Bryozoa
 Calamus bajonado²
 Calamus calamus²
 Calamus leucosteus²
 Haemulon carbonarium²
 Haemulon flavolineatum²
 Haemulon macrostomum²
 Haemulon melanurum²
 Haemulon parra²
 Haemulon plumierii²
 Haemulon sciurus²
 Haemulon sp.²
 Haemulon striatum²
 Haemulon vittatum²
 Halichoeres bivittatus²
 Halichoeres caudalis²
 Halichoeres cyanocephalus²
 Halichoeres garnoti²
 Halichoeres maculipinna²
 Halichoeres pictus²
 Halichoeres poeyi²
 Halichoeres radiatus²
 Haliclona (Reneira) aquaeductus
 Haliclona (Reniera) tubifera
 Haliclona sp.
 Halimeda spp.¹
 Halisarca sp.
 Halodule wrightii¹
 Halophila decipiens¹
 Halophila engelmannii¹
 Halophila johnsonii¹
 Halophila sp.¹
 Harengula humeralis
 Harengula jaguana
 Helioseris cucullata³
 Hemiemblemaria simula²
 Hemiramphus brasiliensis
 Heteroconger longissimus
 Heteropriacanthus cruentatus²
 Hexacorallia
 Higginsia strigilata
 Hippocampus erectus²
 Hippocampus reidi²
 Hippospongia sp.
 Holacanthus bermudensis²
 Holacanthus ciliaris²
 Holacanthus tricolor²
 Holocentrus adscensionis²
 Holocentrus rufus²
 Hydrozoa
 Hypanus americanus
 Hycleurochilus bermudensis²
 Hypnea¹
 Hypoatherina harringtonensis
 Hypoglossum¹
 Hypoplectrus chlorurus²
 Hypoplectrus gemma²
 Hypoplectrus gummigutta²
 Hypoplectrus guttavarius²
 Hypoplectrus hybrid²
 Hypoplectrus indigo²
 Pseudoplexaura porosa³
 Pseudoplexaura wagnaari³
 Pseudopeneus maculatus²
 Ptereleotris calliura
 Ptereleotris helenae
 Pterocladia sanctarum¹
 Pterogorgia anceps³
 Pterogorgia citrina³
 Pterogorgia guadalupensis³
 Pterois miles²
 Pterois volitans²
 Ptilocaulis sp.
 Rachycentron canadum
 Ramicrusta spp.
 Razorfish sp.²
 Red calcareous branching algae
 Red frondose algae
 Remora remora
 Rhinoptera bonasus
 Rhipocephalus¹
 Rhipocephalus phoenix¹
 Rhipocephalus spp.¹
 Rhodactis osculifera
 Rhomboplites aurorubens²
 Ricordea florida
 Rubble
 Ruppia maritima¹
 Rypticus bistrispinus²
 Rypticus maculatus²
 Rypticus saponaceus²
 Sand-sand
 Sand on hard-bottom
 Sardinella aurita
 Sargassum sp.¹
 Sargassum spp.¹
 Sargocentron bullisi²
 Sargocentron coruscum²
 Sargocentron vexillarium²
 Scartella cristata²
 Scarus coelestinus²
 Scarus coeruleus²
 Scarus guacamaia²
 Scarus iseri²
 Scarus sp.²
 Scarus taeniopterus²
 Scarus vetula²
 Schizothrix calcicola
 Schultzea beta²
 Scianid sp.²
 Scleractinia³
 Scolymia cubensis³
 Scolymia lacera
 Scolymia sp.³
 Scolymia spp.³
 Scomberomorus cavalla
 Scomberomorus maculatus
 Scomberomorus regalis

Calamus nodosus ²	Hypoplectrus nigricans ²	Scopalina ruetzleri
Calamus penna ²	Hypoplectrus puella ²	Scorpaena plumieri ²
Calamus proridens ²	Hypoplectrus sp. ²	Scorpaenodes caribbaeus ²
Calcareous green algae ¹	Hypoplectrus tann ²	Selachii
Callionymus bairdi ²	Hypoplectrus unicolor ²	Selene vomer ²
Callyspongia (Callyspongia) fallax	Hyporthodus flavolimbatus ²	Seriola dumerili ²
Callyspongia (Cladochalina) aculeata	Hyporthodus niveatus ²	Seriola rivoliana ²
Callyspongia (Cladochalina) plicifera	Hyrtios violaceus	Seriola sp. ²
Callyspongia (Cladochalina) tenerrima	Iciligorgia schrammi ³	Seriola zonata ²
Calyx podatypa	Iotrochota birotulata	Serranid sp. ²
Cantherhines macrocerus ²	Ircinia campana	Serranus annularis ²
Cantherhines pullus ²	Ircinia felix	Serranus baldwini ²
Canthidermis sufflamen ²	Ircinia strobilina	Serranus phoebe ²
Canthigaster rostrata ²	Ircinia variabilis	Serranus subligarius ²
Caranx bartholomaei ²	Isophyllia rigida ³	Serranus tabacarius ²
Caranx crysos ²	Isophyllia sinuosa ³	Serranus tigrinus ²
Caranx hippos ²	Isophyllia sp.	Serranus tortugarum ²
Caranx latus ²	Istiophorus platypterus	Siderastrea radians ³
Caranx lugubris ²	Jania spp. ¹	Siderastrea siderea ³
Caranx ruber ²	Jenkinsia sp.	Siderastrea sp.
Caranx sp. ²	Kallymenia spp.	Siderastrea spp.
Carcharhinus falciformis	Kyphosus sectatrix ²	Silt on hard-bottom
Carcharhinus leucas	Labrisomidae sp. ²	Siphonodictyon coralliphagum
Carcharhinus limbatus	Labrisomus nuchipinnis ²	Siphonodictyon siphonum
Carcharhinus obscurus	Lachnolaimus maximus ²	Snapper sp. ²
Carcharhinus perezii	Lactophrys bicaudalis ²	Solenastrea bournoni ³
Carcharhinus plumbeus	Lactophrys trigonus ²	Solenastrea hyades ³
Caulerpa spp. ¹	Lactophrys triqueter ²	Solenastrea sp.
Centroceras sp. ¹	Lagodon rhomboides ²	Solenastrea spp. ³
Centropomus undecimalis	Laurencia spp. ¹	Sparidae sp. ²
Centropristis ocyurus ²	Lebrunia neglecta	Sparisoma atomarium ²
Centropristis striata ²	Liagora spp.	Sparisoma aurofrenatum ²
Centropyge argi ²	Liopropoma eukrines ²	Sparisoma chrysopterum ²
Cephalopholis cruentata ²	Liopropoma mowbrayi ²	Sparisoma radians ²
Cephalopholis fulva ²	Liopropoma rubre ²	Sparisoma rubripinne ²
Ceramium ¹	Lobophora spp.	Sparisoma sp. ²
Chaenopsis limbaughi ²	Lutjanus analis ²	Sparisoma viride ²
Chaetodipterus faber	Lutjanus apodus ²	Spermothamnion ¹
Chaetodon capistratus ²	Lutjanus buccanella ²	Spheciospongia vesparium
Chaetodon ocellatus ²	Lutjanus campechanus ²	Sphoeroides ²
Chaetodon sedentarius ²	Lutjanus cyanopterus ²	Sphoeroides nephelus ²
Chaetodon striatus ²	Lutjanus griseus ²	Sphoeroides spengleri ²
Chaetomorpha linum ¹	Lutjanus jocu ²	Sphoeroides testudineus ²
Champia parvula ¹	Lutjanus mahogoni ²	Sphyraena barracuda ²
Chara spp. ¹	Lutjanus synagris ²	Sphyraena guachancho
Chilomycterus antennatus ²	Macroalgae	Sphyraena picudilla
Chilomycterus reticulatus ²	Madracis aurentra ³	Sphyrna lewini
Chilomycterus schoepfii ²	Madracis carmabi ³	Sphyrna mokarran
Chloroscombrus chrysurus ²	Madracis decactis ³	Sphyrna tiburo
Chondria capillaris ¹	Madracis formosa ³	Spirastrella coccinea
Chondria spp. ¹	Madracis myriaster ³	Spirastrella mollis
Chondrilla nucula	Madracis senaria ³	Spongia sp.
Chondrosia sp.	Madracis sp.	Spyridia filamentosa ¹
Chriodorus atherinoides	Madracis spp. ³	Squirrelfish sp. ²
Chromis enchrysurus ²	Malacanthus plumieri	Stegastes adustus ²
Chromis insolata ²	Malacoctenus aurolineatus ²	Stegastes diencaeus ²
Chromis multilineata ²	Malacoctenus gilli ²	Stegastes leucostictus ²

<i>Chromis scotti</i> ²	<i>Malacoctenus macropus</i> ²	<i>Stegastes partitus</i> ²
<i>Cinachyra</i> sp.	<i>Malacoctenus triangulatus</i> ²	<i>Stegastes planifrons</i> ²
<i>Cladocephalus</i> ¹	<i>Malacoctenus versicolor</i> ²	<i>Stegastes</i> sp. ²
<i>Cladocora arbuscula</i> ³	<i>Manicina areolata</i> ³	<i>Stegastes variabilis</i> ²
<i>Cladophora</i> ¹	<i>Meandrina jacksoni</i>	<i>Stephanocoenia intersepta</i> ³
<i>Clathria</i> (<i>Thalysias</i>) <i>venosa</i>	<i>Meandrina meandrites</i> ³	<i>Stephanolepis hispida</i> ²
<i>Clathria</i> (<i>Thalysias</i>) <i>virgultosa</i>	<i>Megalops atlanticus</i>	<i>Stichodactyla helianthus</i>
<i>Clathria</i> sp.	<i>Melichthys niger</i> ²	<i>Strongylacidon</i> sp.
<i>Clepticus parrae</i> ²	<i>Menidia</i> sp.	<i>Strongylura notata</i> ²
<i>Cliona caribbaea</i>	<i>Microgobius carri</i> ²	<i>Strongylura timucu</i>
<i>Cliona delitrix</i>	<i>Microgobius microlepis</i> ²	<i>Stygnobrotula latebricola</i>
<i>Cliona</i> sp.	<i>Microspathodon chrysurus</i> ²	<i>Stypopodium</i> spp.
<i>Cliona</i> spp.	<i>Millepora alcornis</i> ³	Substrate
<i>Cliona varians</i>	<i>Millepora complanata</i> ³	<i>Syacium micrurum</i> ²
<i>Colpophyllia natans</i> ³	<i>Millepora</i> spp. ³	<i>Syngnathus scovelli</i> ²
<i>Colpophyllia</i> spp. ³	<i>Mobula birostris</i>	<i>Syngnathus</i> sp. ²
<i>Condylactis gigantea</i>	<i>Monacanthus ciliatus</i> ²	<i>Synodus foetens</i> ²
<i>Corallimorpharians</i>	<i>Monacanthus tuckeri</i> ²	<i>Synodus intermedius</i> ²
<i>Coralliophila erosa</i>	<i>Monanchora arbuscula</i>	<i>Synodus synodus</i> ²
<i>Coryphopterus dicrus</i> ²	<i>Montastraea cavernosa</i> ³	<i>Syringodium filiforme</i> ¹
<i>Coryphopterus eidolon</i> ²	<i>Mulloidichthys martinicus</i> ²	<i>Tectitethya crypta</i>
<i>Coryphopterus glaucofraenum</i> ²	<i>Muraena retifera</i> ²	<i>Tedania</i> (<i>Tedania</i>) <i>ignis</i>
<i>Coryphopterus lipernes</i> ²	<i>Muricea atlantica</i> ³	<i>Tethya diploderma</i>
<i>Coryphopterus personatus</i> ²	<i>Muricea elongata</i> ³	<i>Thalassia testudinum</i> ¹
<i>Coryphopterus punctipictophorus</i> ²	<i>Muricea laxa</i> ³	<i>Thalassoma bifasciatum</i> ²
<i>Coryphopterus</i> sp. ²	<i>Muricea muricata</i> ³	Thick turf
<i>Cribrochalina vasculum</i>	<i>Muricea pinnata</i> ³	<i>Tigrigobius macrodon</i> ²
<i>Crustose coralline algae</i> ¹	<i>Muriceopsis flavida</i> ³	<i>Tigrigobius saucrus</i> ²
<i>Cryptotomus roseus</i> ²	<i>Mussa angulosa</i> ³	Total brown algae ¹
<i>Ctenogobius saepepallens</i> ²	<i>Mycale</i> (<i>Mycale</i>) <i>laevis</i>	Total calcareous green algae ¹
<i>Cyanobacteria</i>	<i>Mycale</i> sp.	Total green algae ¹
<i>Cyanophyta</i> spp.	<i>Mycetophyllia aliciae</i> ³	Total macroalgae ¹
<i>Dactylopterus volitans</i>	<i>Mycetophyllia danaana</i> ³	Total other green algae ¹
<i>Dasya</i> sp. ¹	<i>Mycetophyllia ferox</i> ³	Total other red algae ¹
<i>Dasycladus</i> ¹	<i>Mycetophyllia lamarckiana</i> ³	Total SAV ¹
<i>Decapterus macarellus</i> ²	<i>Mycetophyllia</i> sp.	Total seagrass ¹
<i>Decapterus punctatus</i> ²	<i>Mycetophyllia</i> spp. ³	<i>Trachinotus falcatus</i> ²
<i>Decapterus</i> sp. ²	<i>Mycteroperca acutirostris</i> ²	<i>Trachinotus goodei</i> ²
<i>Dendrogyra cylindrus</i> ³	<i>Mycteroperca bonaci</i> ²	<i>Trachurus lathami</i> ²
<i>Derbesia</i> ¹	<i>Mycteroperca interstitialis</i> ²	<i>Trachyteleia hispida</i>
<i>Dermatolepis inermis</i> ²	<i>Mycteroperca microlepis</i> ²	<i>Tunicata</i>
<i>Desmapsamma anchorata</i>	<i>Mycteroperca phenax</i> ²	<i>Turbinaria turbinata</i>
<i>Diadema antillarum</i>	<i>Mycteroperca tigris</i> ²	Turf algae free of sediment
<i>Dichocoenia stokesii</i> ³	<i>Mycteroperca venenosa</i> ²	Turf algae with sediment
<i>Dictyosphaeria</i> ¹	<i>Myrichthys breviceps</i>	<i>Tylosurus crocodilus</i>
<i>Dictyota</i> spp. ¹	<i>Myrichthys ocellatus</i>	<i>Udotea</i> ¹
<i>Diodon holocanthus</i> ²	<i>Myripristis jacobus</i> ²	<i>Udotea</i> spp. ¹
<i>Diodon hystrix</i> ²	<i>Narcine bancroftii</i>	<i>Ulaema lefroyi</i>
<i>Diodon</i> sp. ²	<i>Needlefish</i> sp.	<i>Ulva</i> sp. ¹
<i>Diplastrella megastellata</i>	<i>Negaprion brevirostris</i>	<i>Umbrina coroides</i> ²
<i>Diplectrum formosum</i> ²	<i>Neofibularia nolitangere</i>	<i>Undaria</i> sp.
<i>Diplodus argenteus</i> ²	<i>Neomeris</i> ¹	Unidentified mangrove
<i>Diplodus holbrookii</i> ²	<i>Neoniphon marianus</i> ²	Unidentified species
<i>Diploria labyrinthiformis</i> ³	<i>Neopetrosia carbonaria</i>	Unknown barrel sponge
<i>Diploria</i> sp.	<i>Nes longus</i> ²	Unknown black smooth encrusting sponge
<i>Diploria</i> spp. ³	<i>Nicholsina usta</i> ²	Unknown bowling ball sponge
<i>Discosoma carlgreni</i>	<i>Niphates amorpha</i>	Unknown brown encrusting sponge

Doratonotus megalepis ²	Niphates digitalis	Unknown brown smooth sponge
Dragmacidon lunaecharta	Niphates erecta	Unknown brown tube sponge
Drift red algae ¹	Oceanapia peltata	Unknown brown vein sponge
Dysidea etheria	Octocorallia ³	Unknown encrusting sponge
Dysidea fragilis	Oculina diffusa ³	Unknown finger/branching sponge
Dysidea janiae	Oculina robusta ³	Unknown green encrusting sponge
Echeneis naucrates	Oculina sp.	Unknown olive sponge
Echeneis neucratoides	Ocyurus chrysurus ²	Unknown orange encrusting sponge
Echinoidea	Odontoscion dentex ²	Unknown orange massive sponge
Ectyoplasia ferox	Ogcocephalus nasutus ²	Unknown pink lumpy sponge
Elacatinus dilepis ²	Ogcocephalus sp.	Unknown red encrusting sponge
Elacatinus evelynae ²	Oligoplites saurus ²	Unknown red lumpy tube sponge
Elacatinus horsti ²	Ophioblennius macclurei ²	Unknown red squishy sponge
Elacatinus oceanops ²	Opistognathus aurifrons	Unknown tube sponge
Elacatinus randalli ²	Opistognathus macrognathus	Upeneus parvus ²
Elacatinus xanthiprora ²	Opistognathus sp.	Urobatis jamaicensis
Elagatis bipinnulata ²	Opistognathus whitehursti	Valonia ¹
Elops saurus	Opsanus tau ²	Verongula gigantea
Emblemaria pandionis ²	Orbicella annularis ³	Verongula reisiwigi
Emblemariopsis bahamensis ²	Orbicella faveolata ³	Verongula rigida
Emmelichthyops atlanticus ²	Orbicella franksi ³	Wrightiella ¹
Enchelycore carychroa ²	Orbicella sp.	Xestospongia muta
Enchelycore nigricans ²	Orthopristis chrysoptera ²	Xyrichtys martinicensis ²
Encrusting gorgonian ³	Other calcareous macroalgae	Xyrichtys novacula ²
Enneanectes altivelis	Other coral	Xyrichtys splendens ²
Enneanectes boehlkei	Other fleshy macroalgae	Zanclus cornutus
Epinephelus adscensionis ²	Other green algae ¹	Zoanthidae
Epinephelus drummondhayi ²	Other red algae ¹	Zoanthids
Epinephelus guttatus ²	Oxyurichthys stigmaliophius ²	Abudefduf saxatilis ²

1 - Submerged Aquatic Vegetation, 2 - Coral Reef - Species Richness, 3 - Coral Reef - Percent Cover

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